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Zhu et al.

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(54) **SYSTEMS AND METHODS FOR
SELECTABLE APPLICATION-SPECIFIC
QUALITY OF SERVICE PARAMETERS IN A
WIRELESS NETWORK**

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This patent is subject to a terminal dis-
claimer.

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H04W 12/00 (2021.01)

(52) **U.S. Cl.**
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(2019.01); **H04W 28/0215** (2013.01); **H04W**
28/0247 (2013.01); **H04W 28/0252** (2013.01)

(58) **Field of Classification Search**

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H04W 12/02; H04W 12/08;

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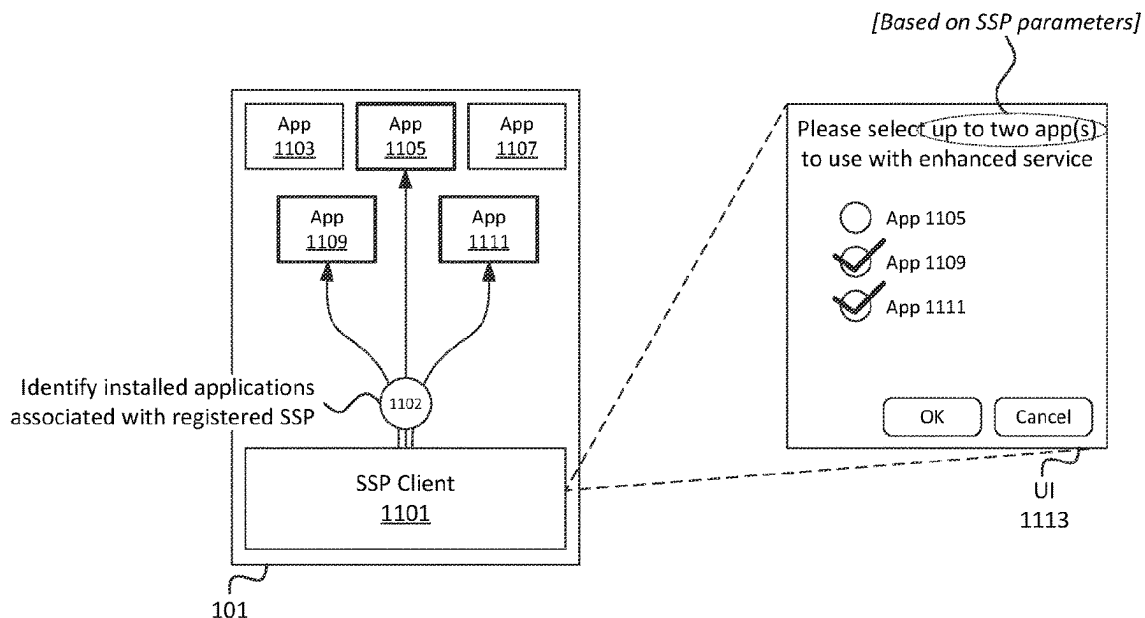
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Primary Examiner — Stephen J Clawson

(57) **ABSTRACT**

A system described herein may apply an authorized set of traffic parameters, such as network slicing parameters, Quality of Service (“QoS”) parameters, etc. to traffic associated with a particular application, without exposing an identifier of the traffic to one or more network devices of a network that handles the traffic. A client associated with a User Equipment (“UE”) may selectively apply traffic parameters to traffic associated with the particular application further based on whether a UE-side authorization of the particular application for the traffic parameters has been performed. The UE may present a user interface via which a user may select whether to authorize or deny the use of traffic parameters, authorized by the network, for the particular application. If the particular application has been authorized by both the network and the user of the UE, the UE may apply the traffic parameters to traffic associated with the particular application.

20 Claims, 21 Drawing Sheets



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See application file for complete search history.

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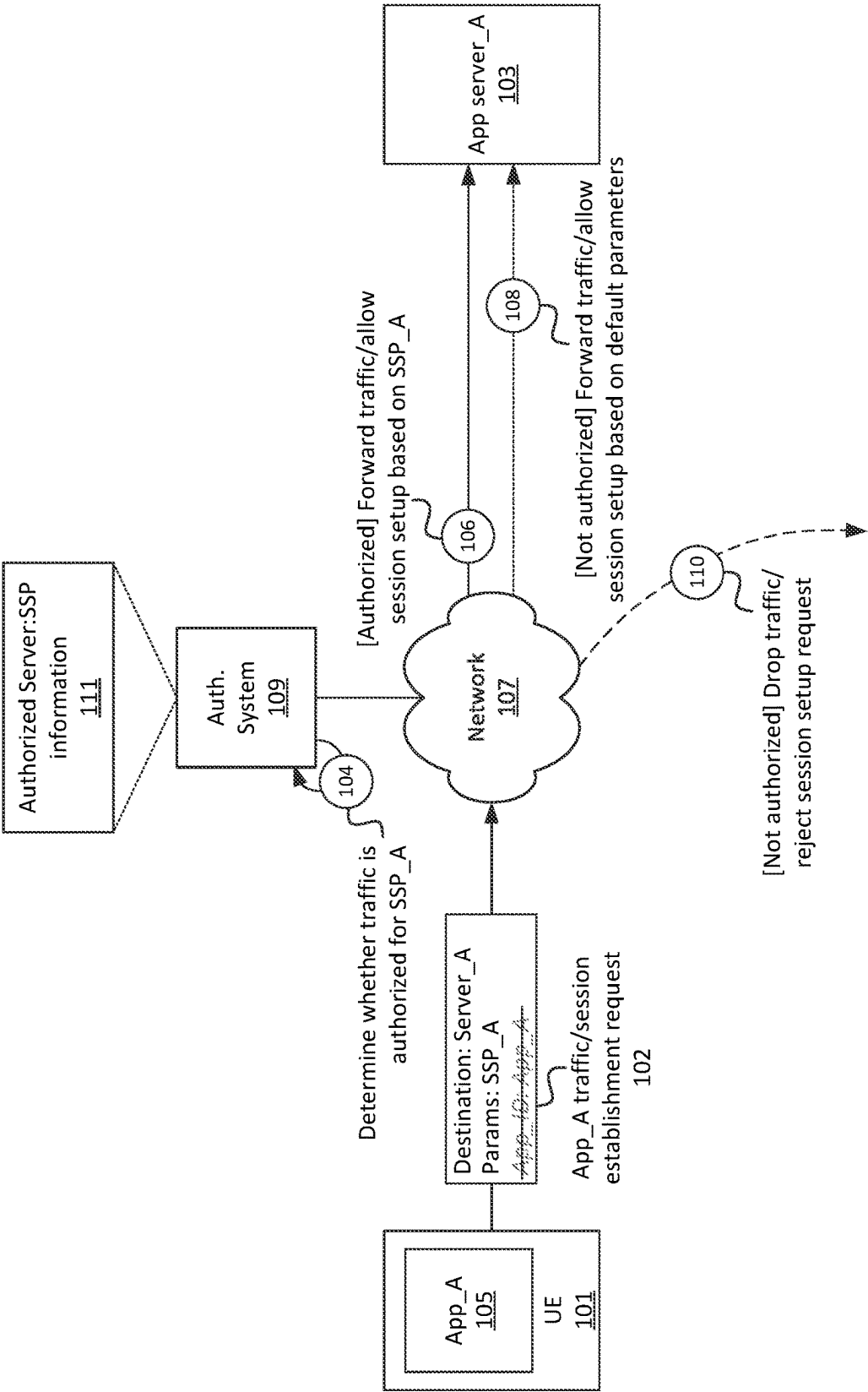


FIG. 1

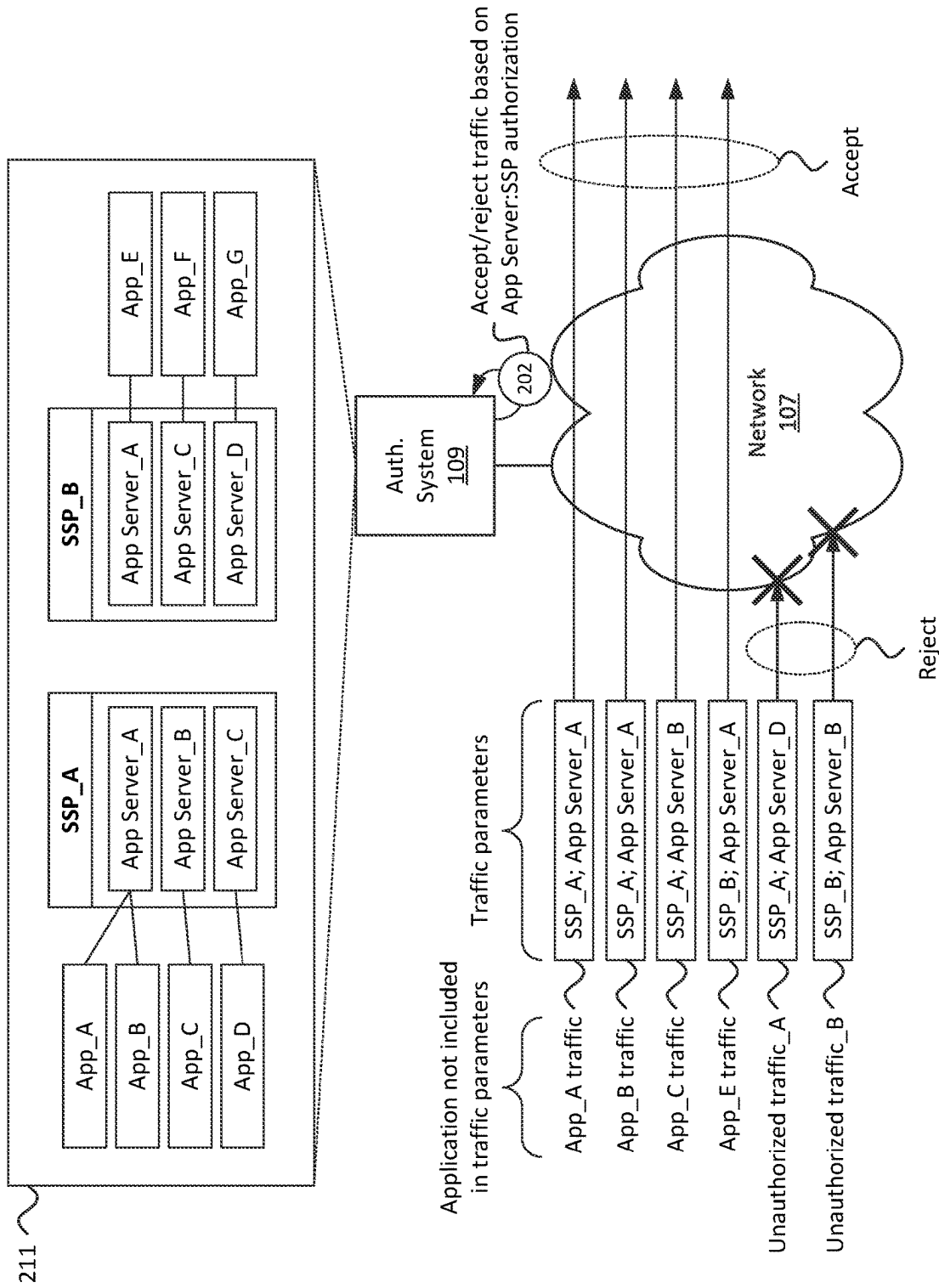


FIG. 2

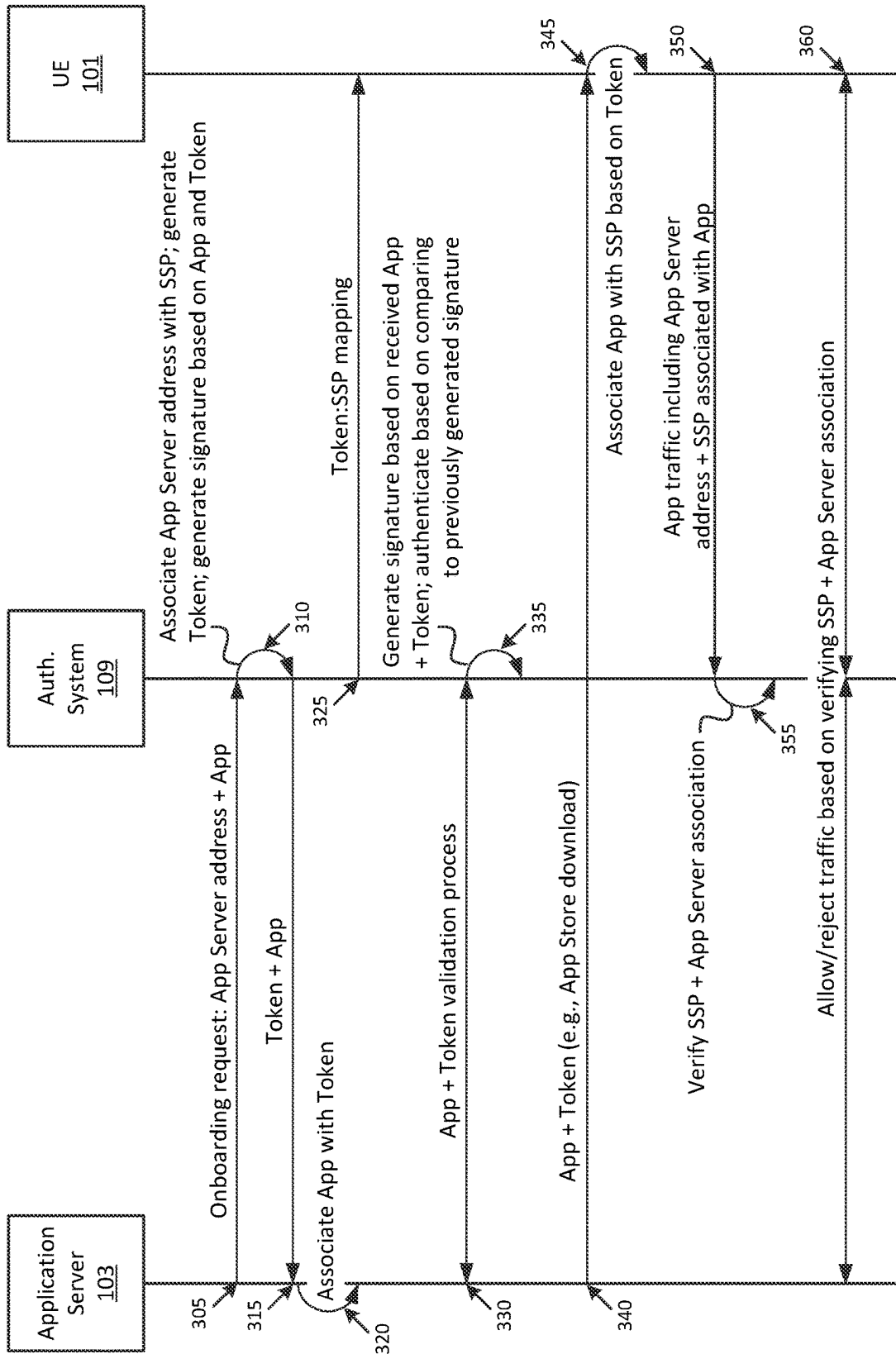
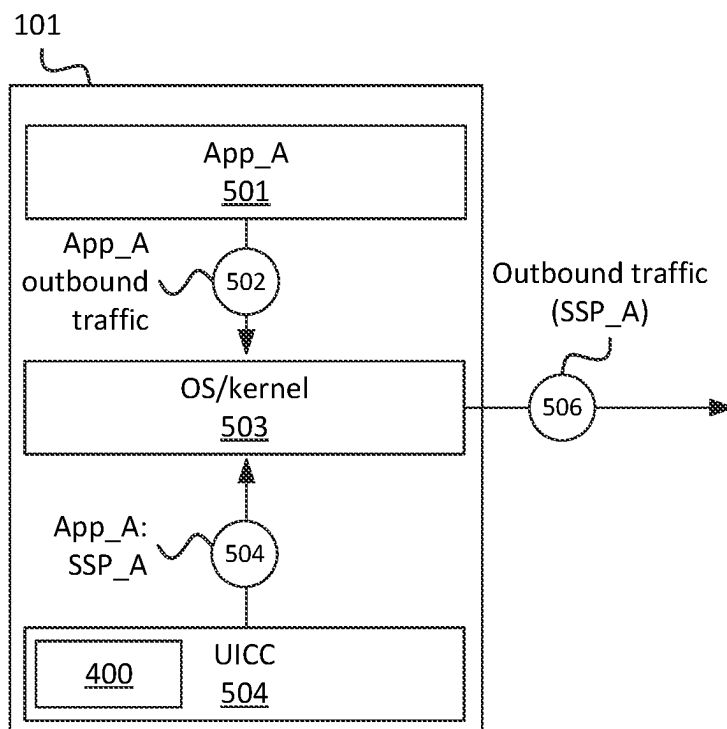
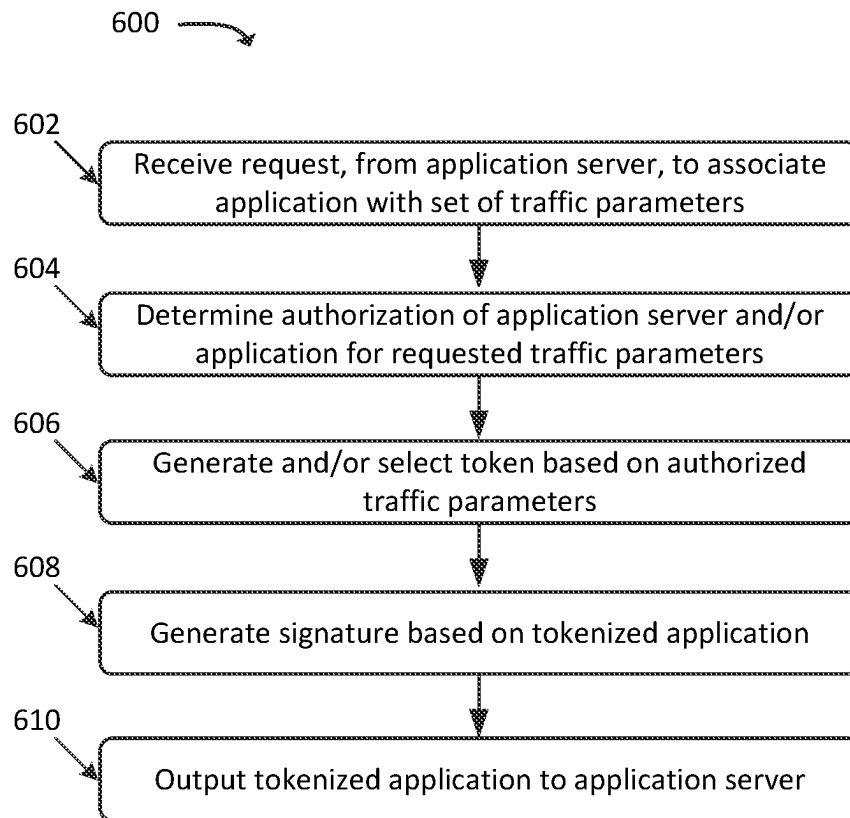


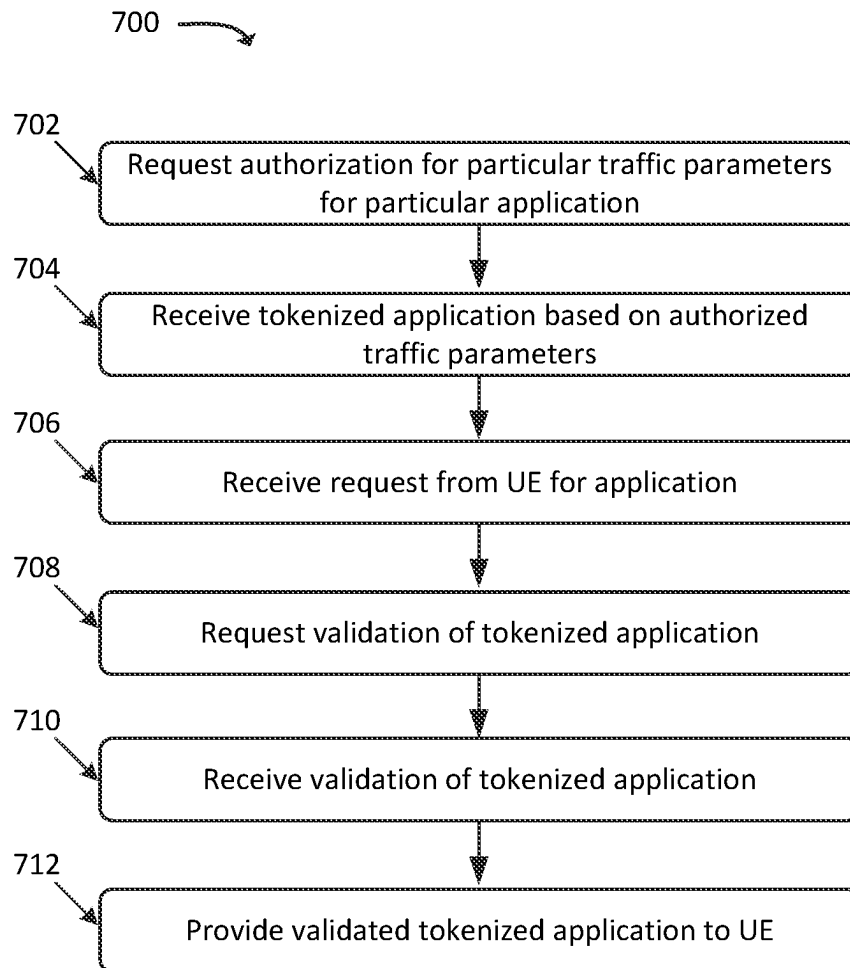
FIG. 3

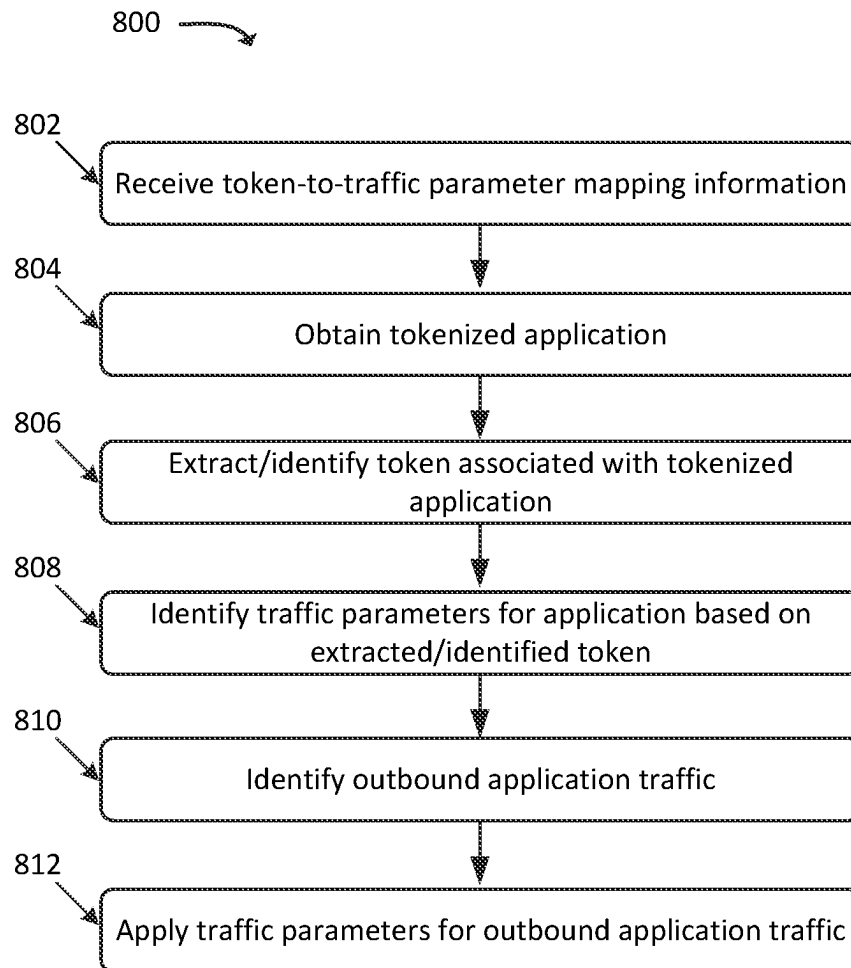
400

Token	SSP
Token_A	SSP_A
Token_B	SSP_B
Token_C	SSP_C
Token_D	SSP_A

FIG. 4**FIG. 5**

**FIG. 6**

**FIG. 7**

**FIG. 8**

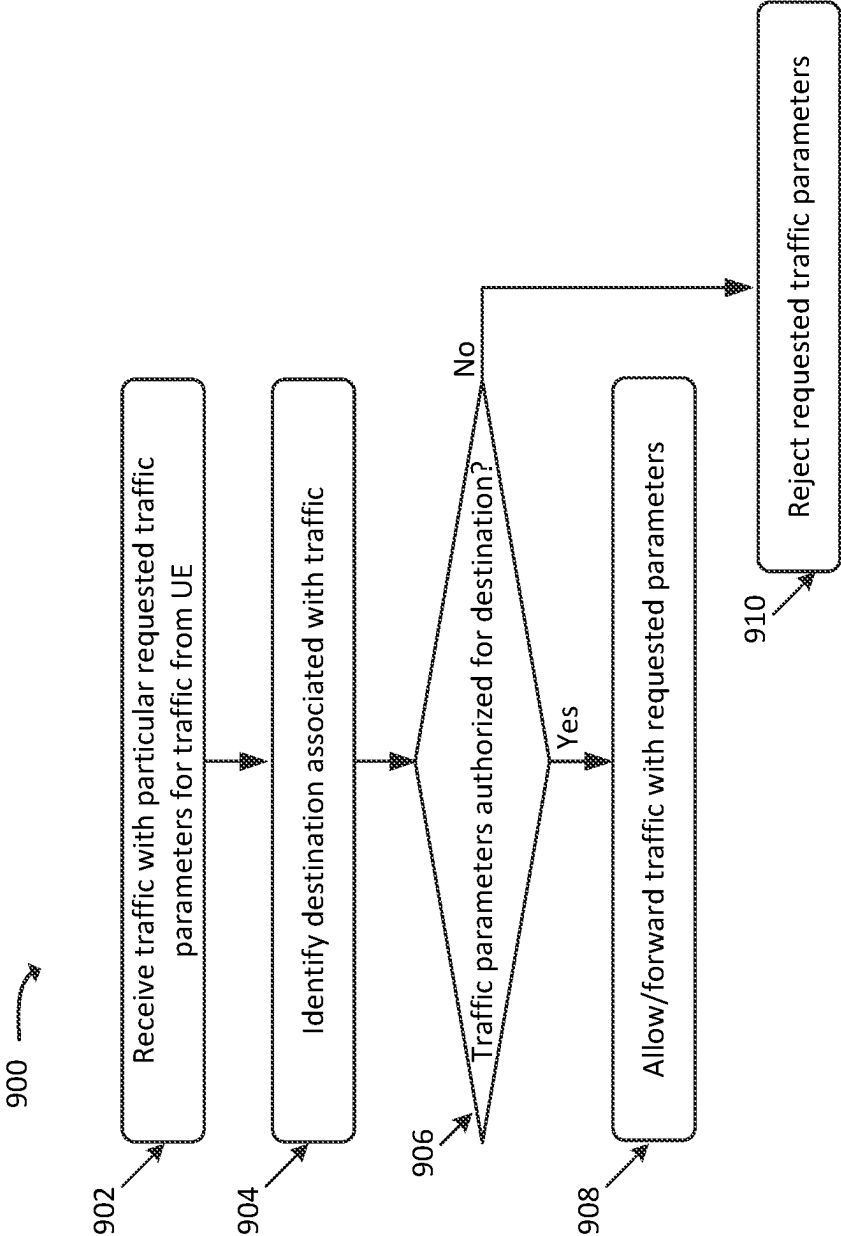
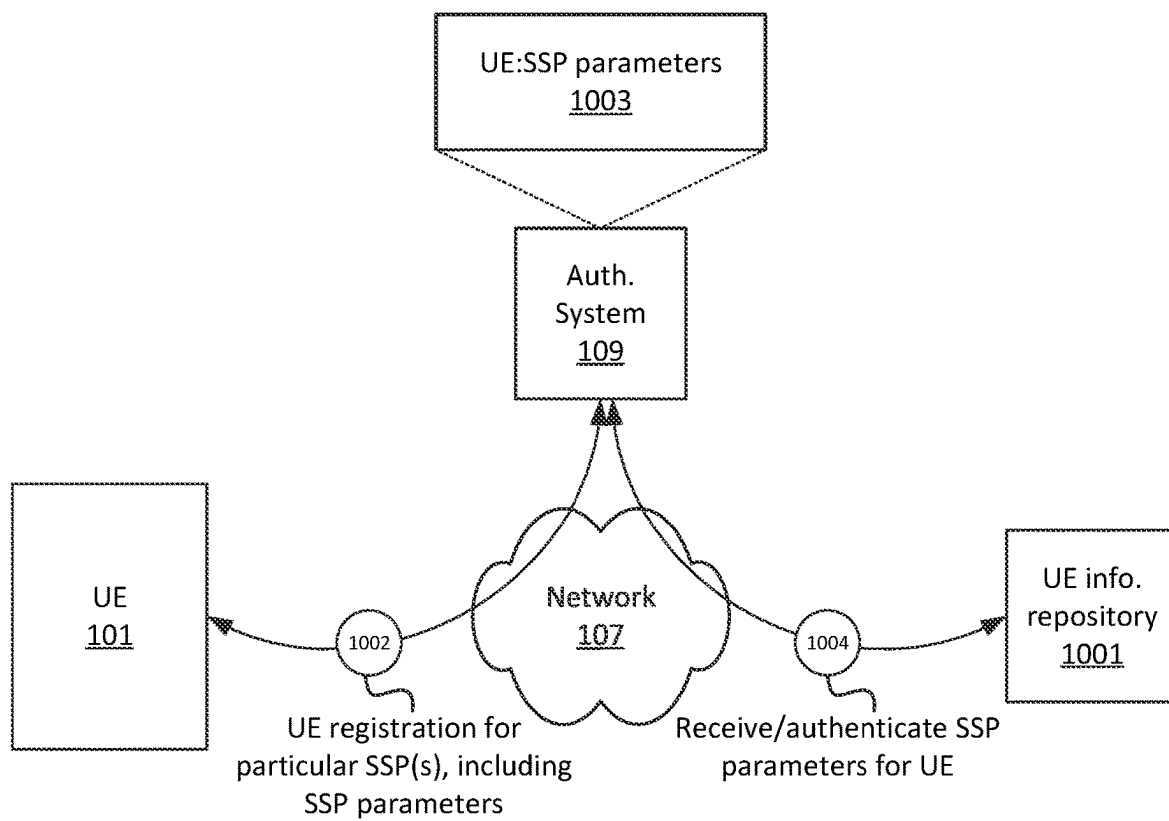


FIG. 9

**FIG. 10**

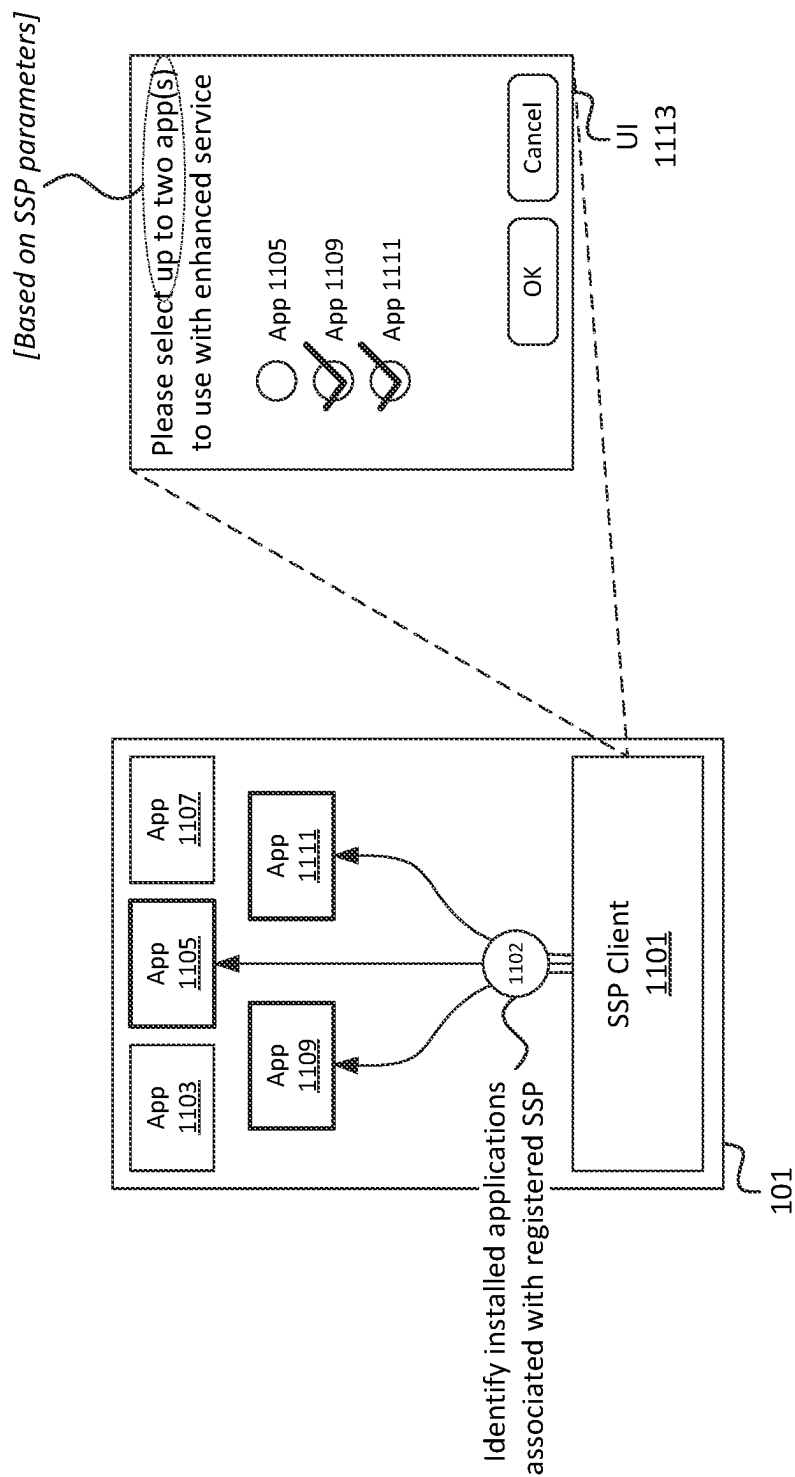


FIG. 11

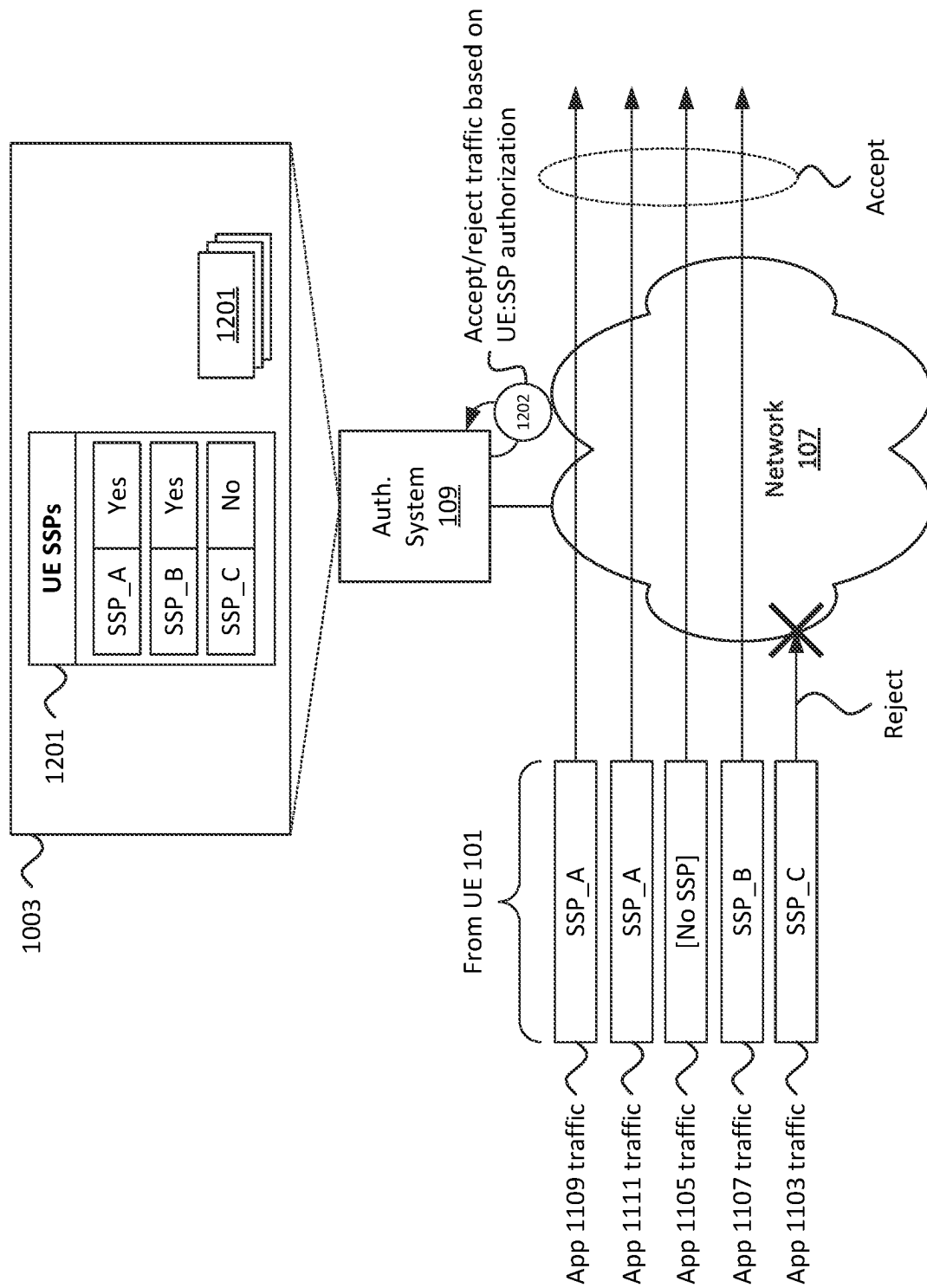


FIG. 12

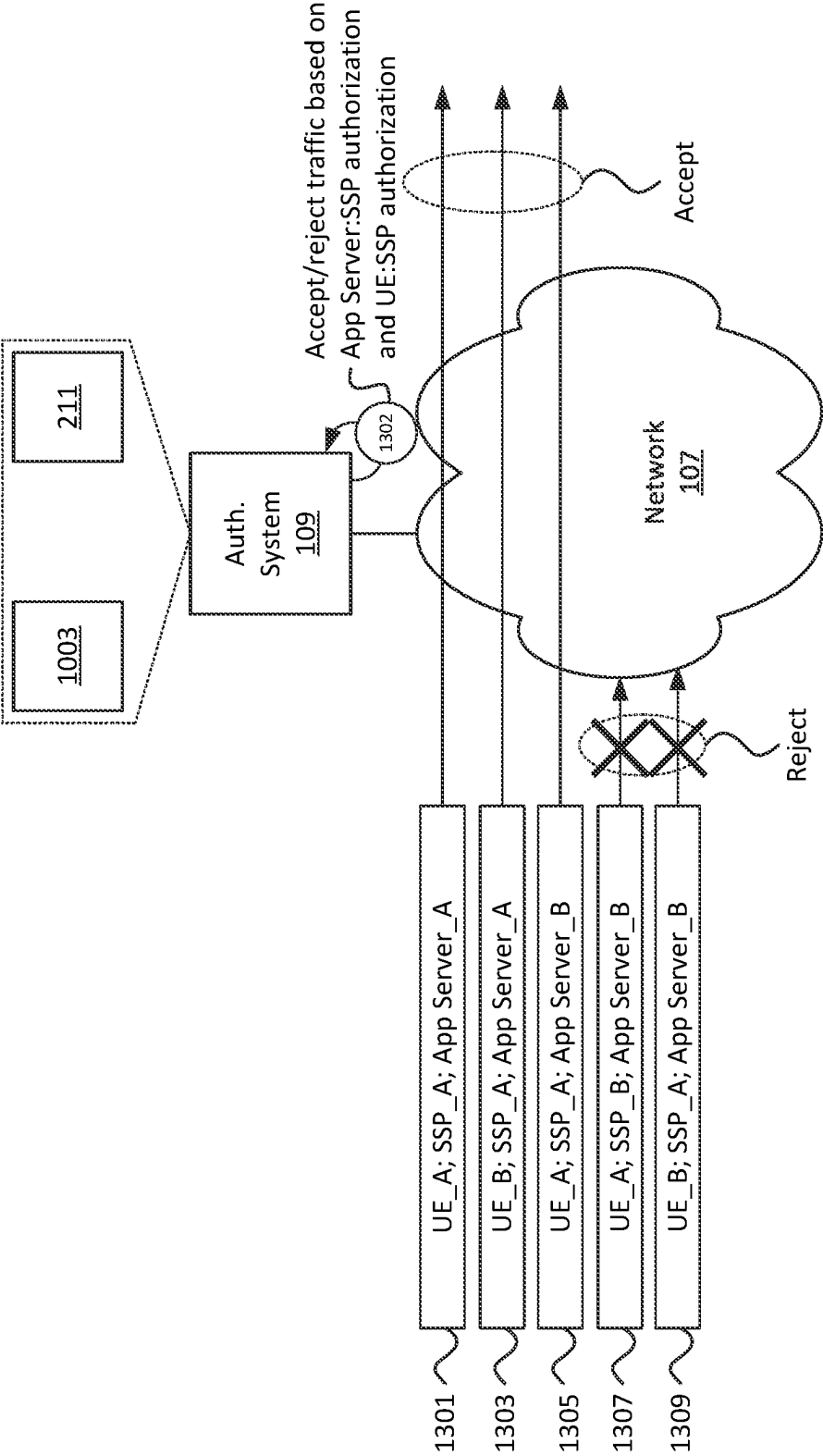


FIG. 13

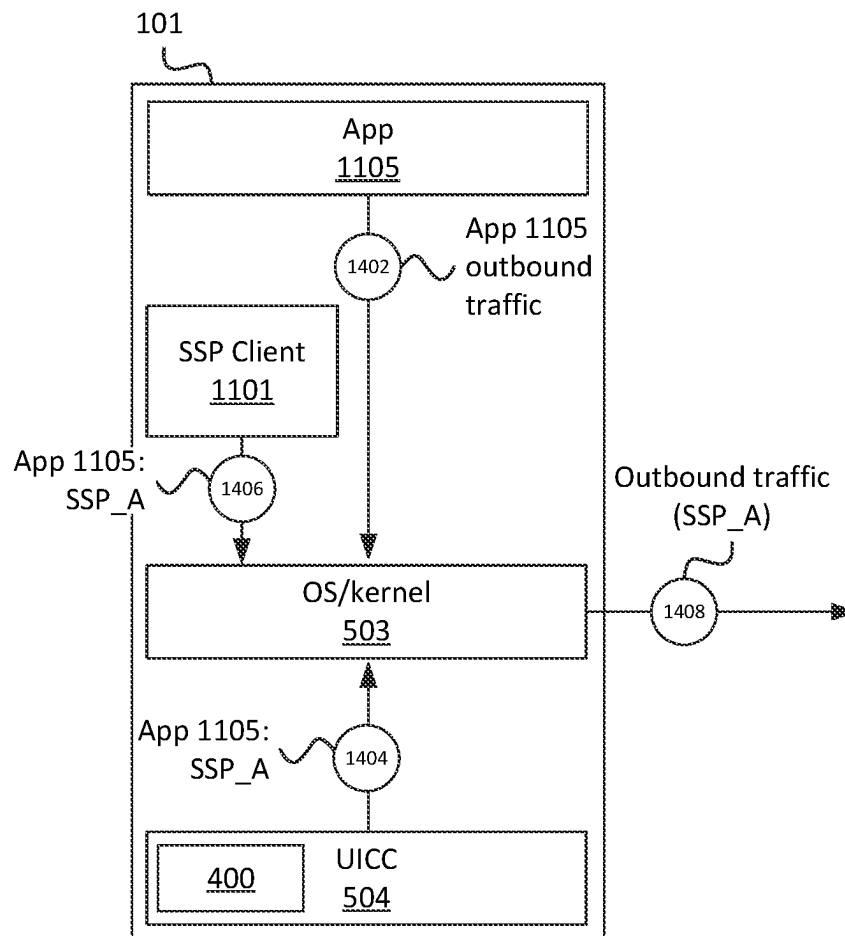
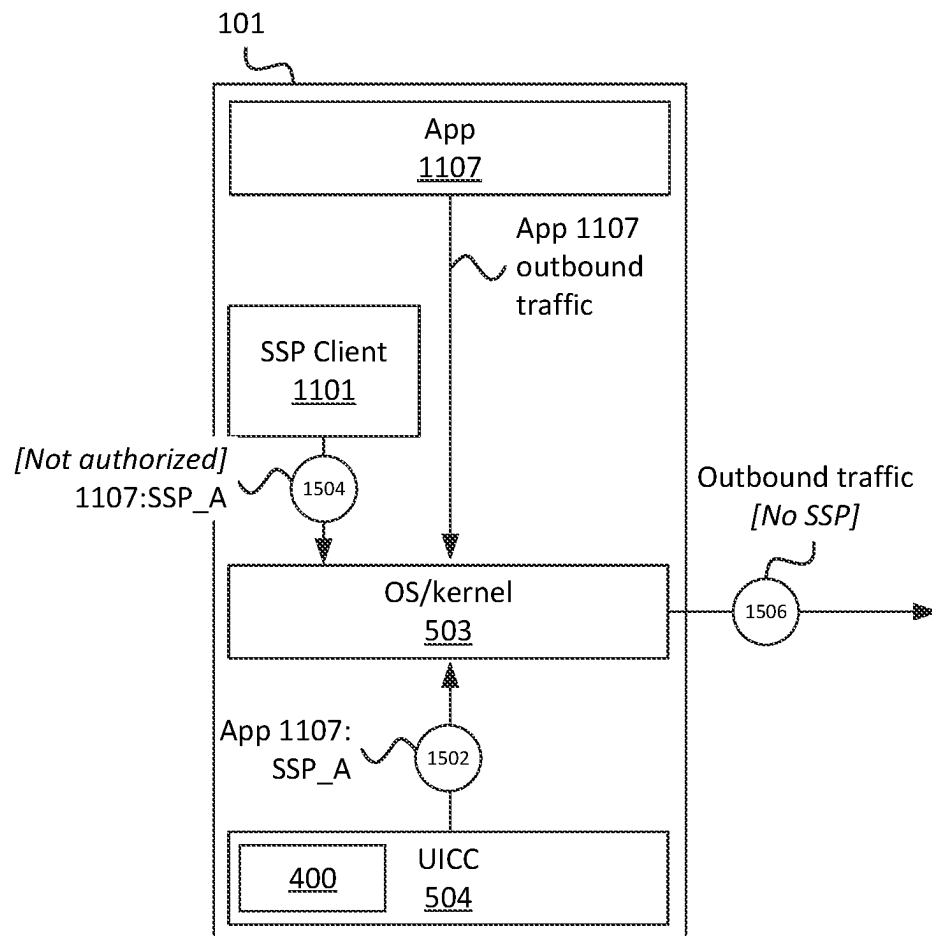
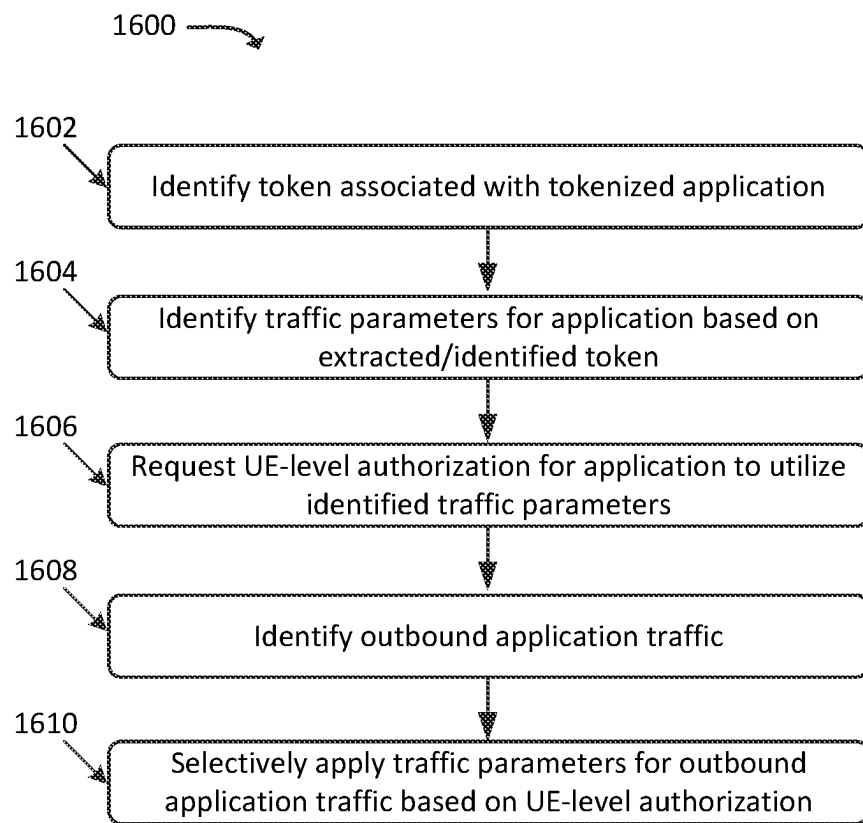


FIG. 14

**FIG. 15**

**FIG. 16**

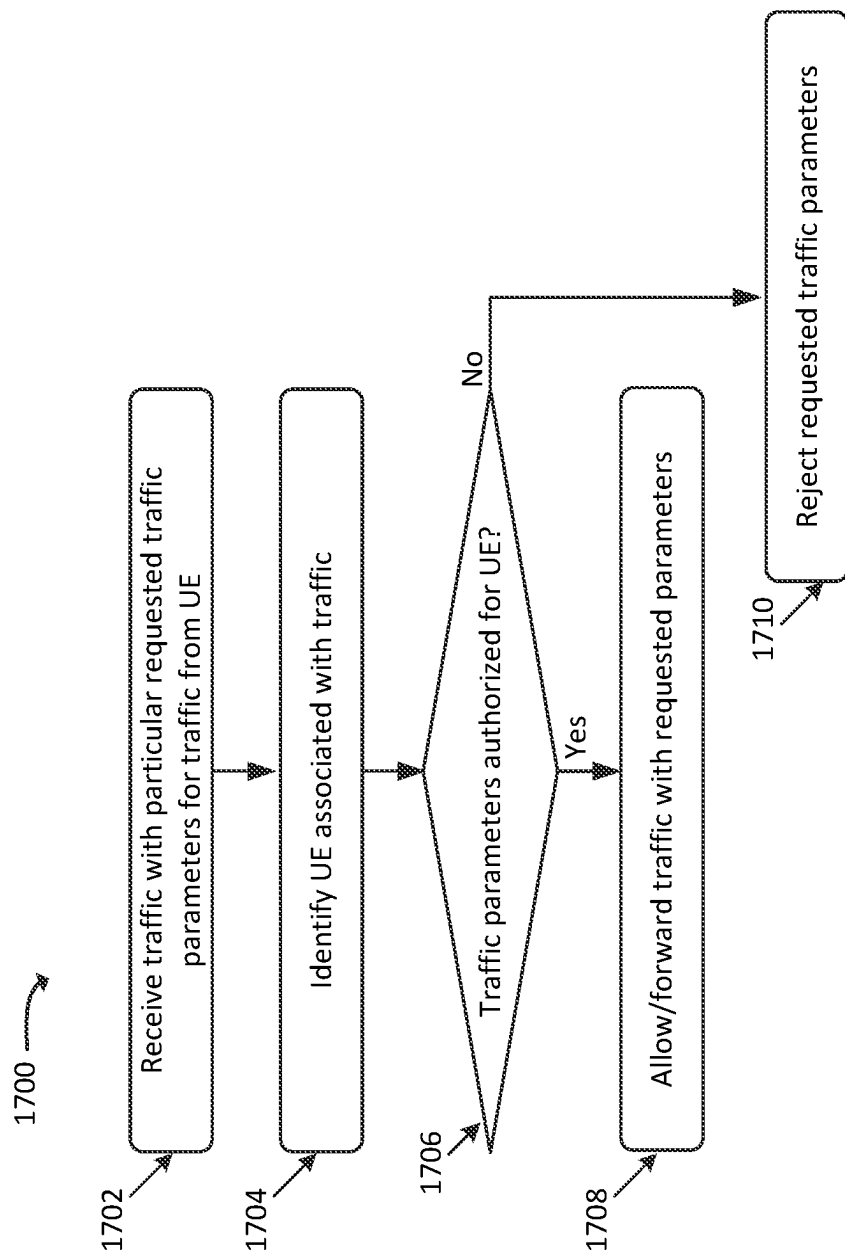
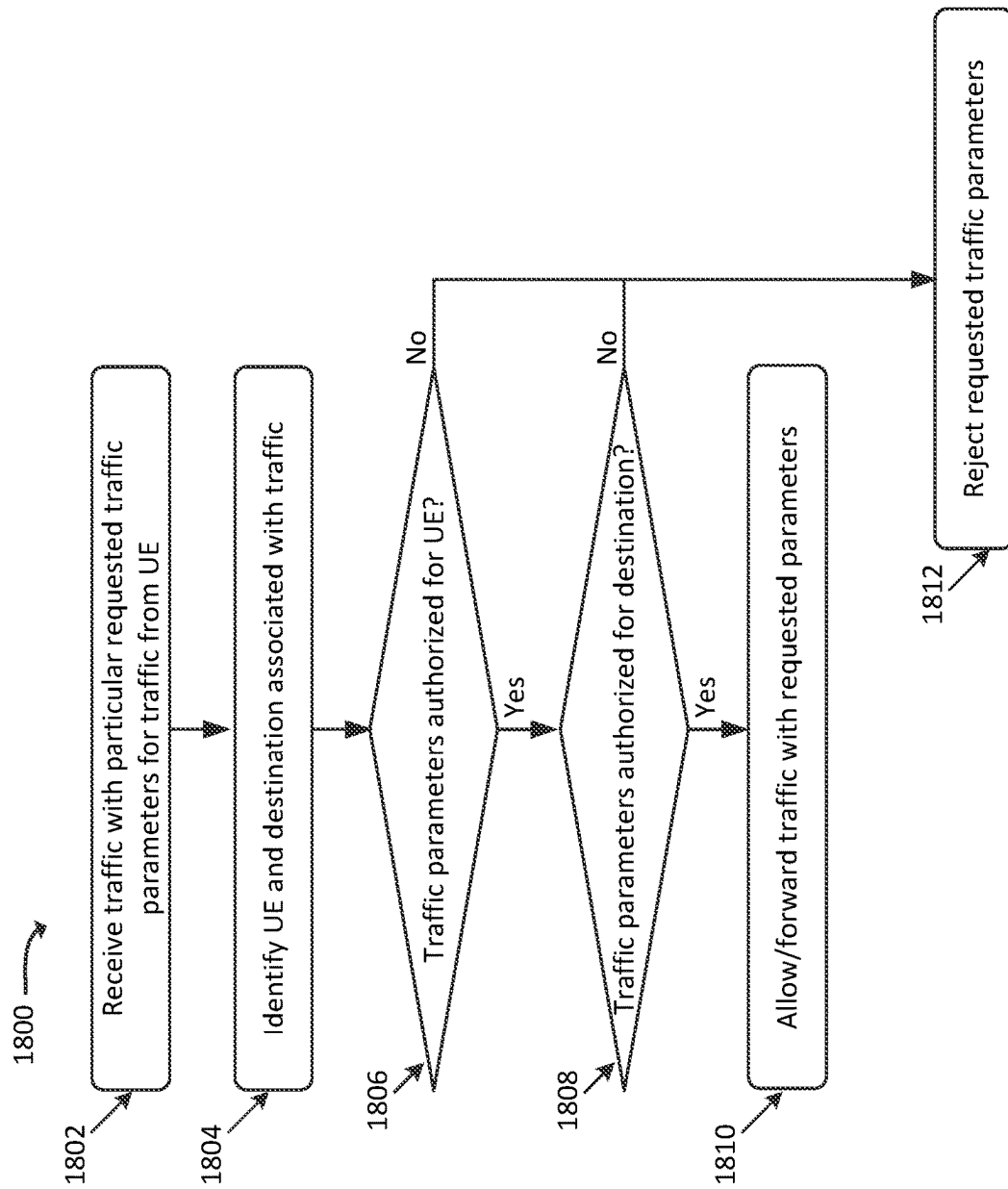


FIG. 17

**FIG. 18**

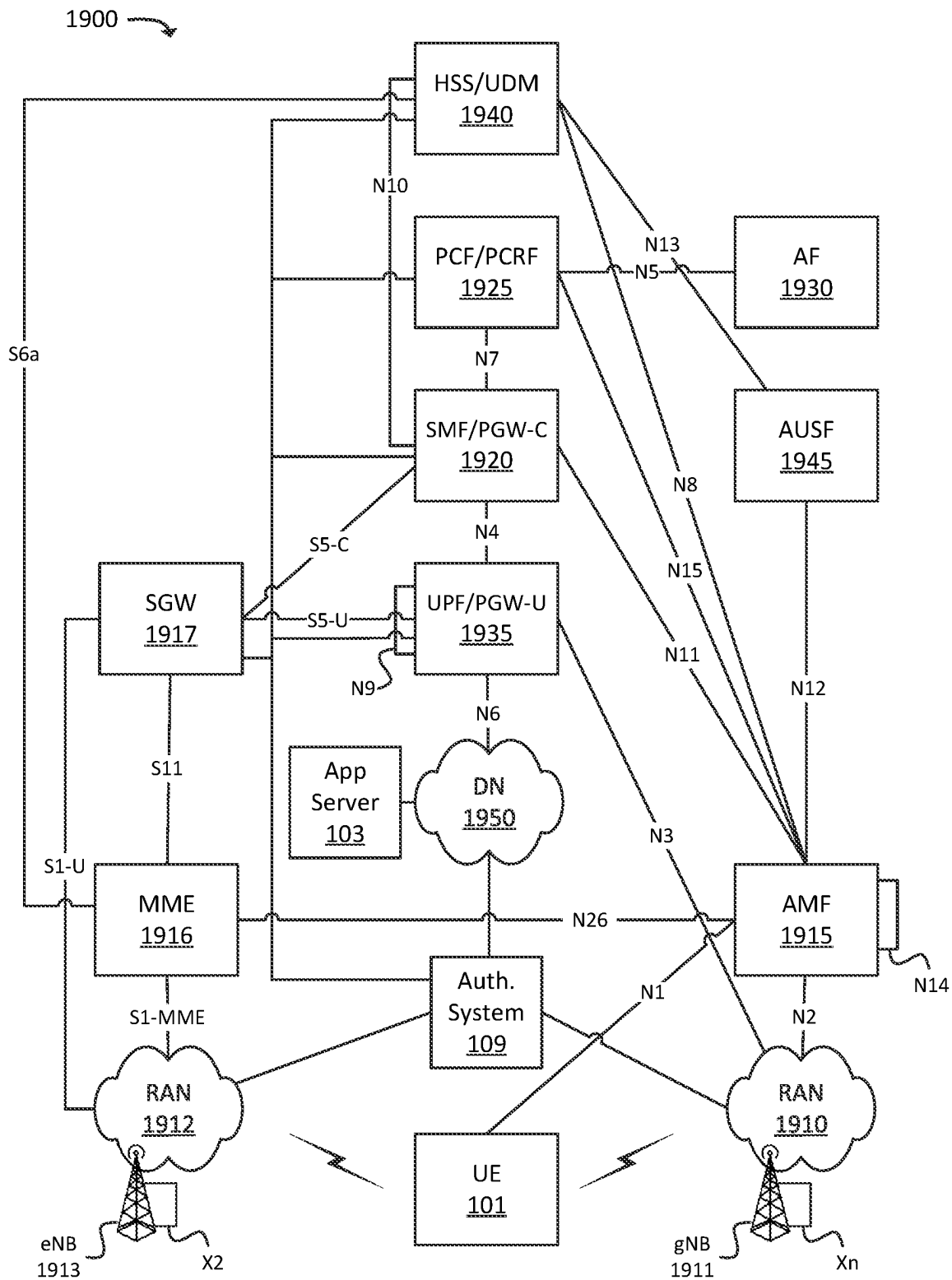
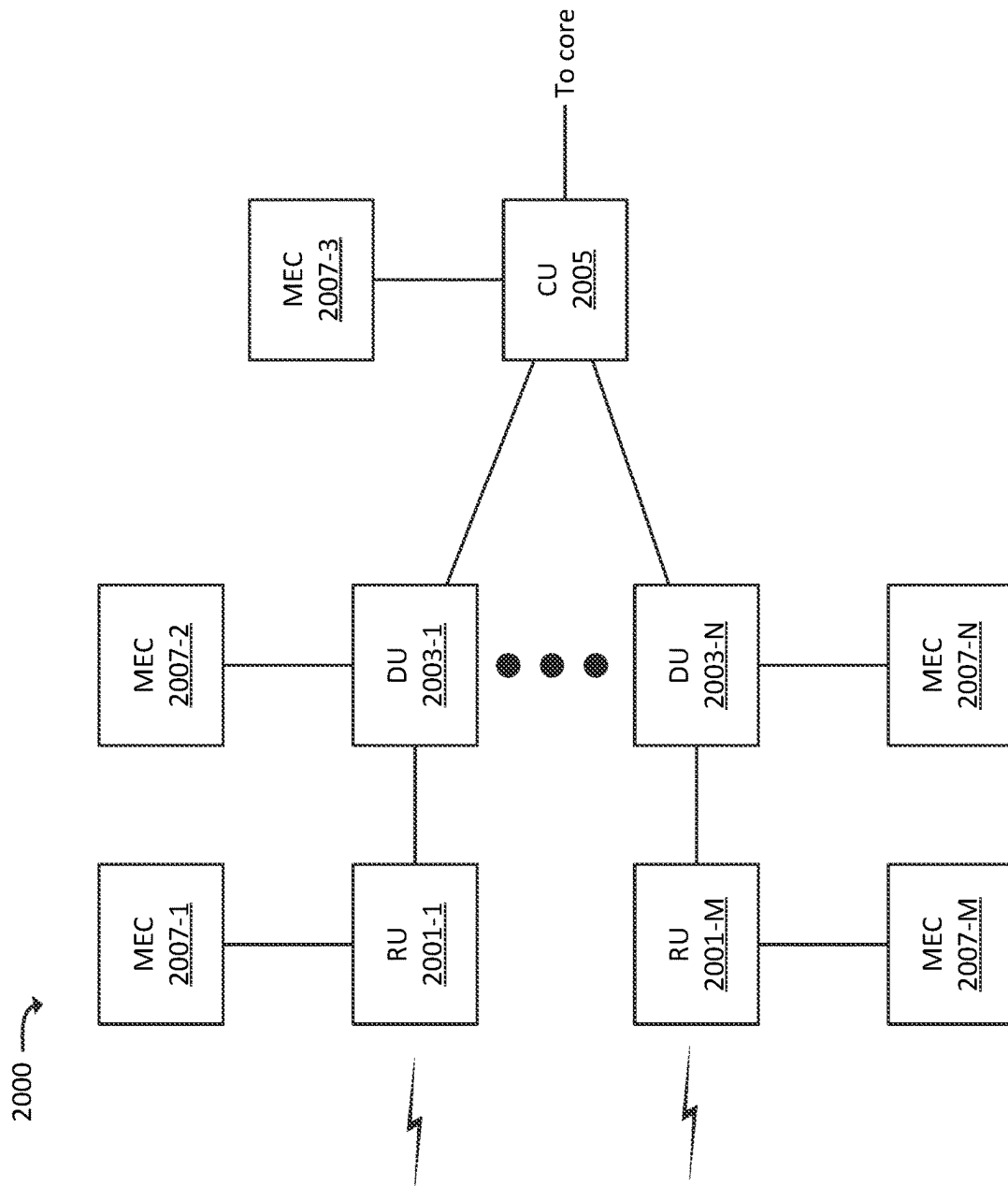


FIG. 19



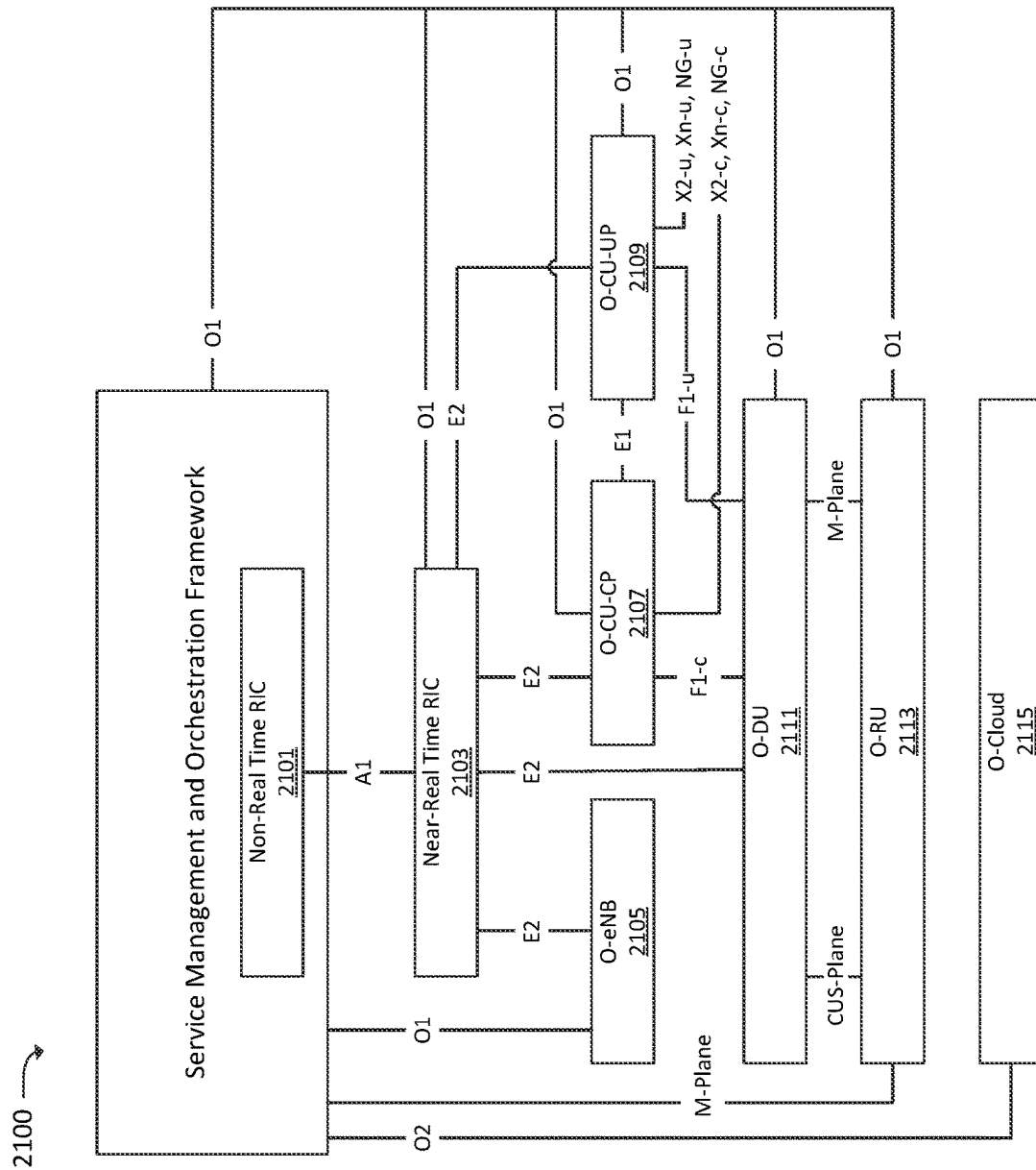
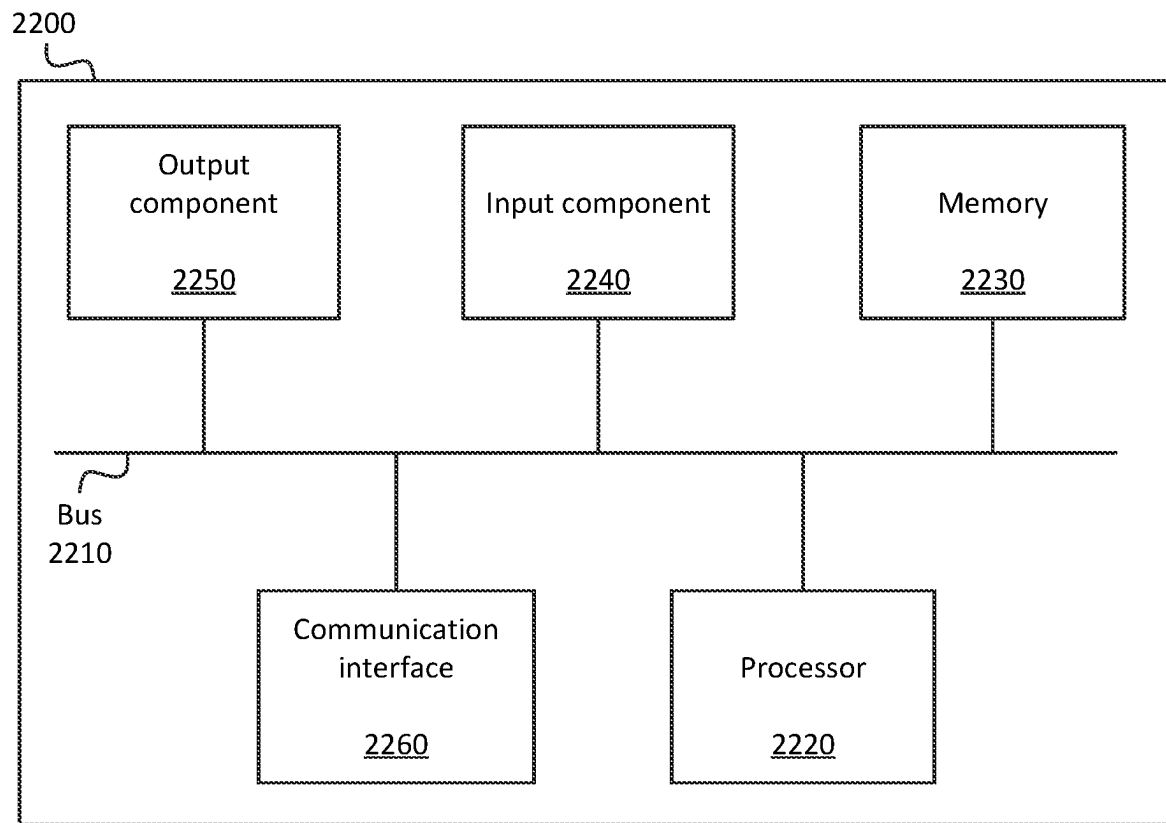


FIG. 21

**FIG. 22**

1

SYSTEMS AND METHODS FOR SELECTABLE APPLICATION-SPECIFIC QUALITY OF SERVICE PARAMETERS IN A WIRELESS NETWORK

CROSS-REFERENCE TO RELATED APPLICATION

This application is a Continuation of U.S. patent application Ser. No. 17/393,565 filed on Aug. 4, 2021, titled “SYSTEMS AND METHODS FOR SELECTABLE APPLICATION-SPECIFIC QUALITY OF SERVICE PARAMETERS IN A WIRELESS NETWORK,” which is a Continuation-in-Part of U.S. patent application Ser. No. 17/194,279, filed on Mar. 7, 2021 and issued as U.S. Pat. No. 11,463,900 on Oct. 4, 2022, titled “NETWORK SLICE POLICY CLIENT FOR CONSUMER DEVICES,” the contents of which are herein incorporated by reference in their entirety.

BACKGROUND

Wireless networks may be used for the wireless transmission of various applications and/or types of traffic, such as voice call traffic, messaging traffic, audio and/or video streaming traffic, file download traffic, and/or other types of traffic. Different applications may have different Quality of Service (“QoS”) requirements, routing rules, or the like. For example, voice call traffic may have lower latency requirements than file download traffic, and file download traffic may have higher throughput requirements than voice call traffic.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates an example overview of one or more embodiments described herein;

FIG. 2 illustrates an example of application-anonymous authorization and/or denial of traffic, in accordance with some embodiments;

FIG. 3 illustrates an example signal flow that may be used to securely provide for application-specific traffic treatment, in accordance with some embodiments;

FIG. 4 illustrates an example data structure that may represent a particular token-to-SSP mapping, in accordance with some embodiments;

FIG. 5 illustrates an example signal flow that may be used to request particular traffic parameters for a given application, in accordance with some embodiments;

FIG. 6 illustrates an example process for authorizing a particular application and application server for a given set of traffic parameters, in accordance with some embodiments;

FIG. 7 illustrates an example process for obtaining authorization for a particular set of traffic parameters for a particular application, in accordance with some embodiments;

FIG. 8 illustrates an example process for identifying and implementing authorized traffic parameters associated with an application, in accordance with some embodiments;

FIG. 9 illustrates an example process for authorizing and/or rejecting traffic with particular traffic parameters based on authorized parameters for a destination of the traffic, in accordance with some embodiments;

FIG. 10 illustrates an example overview of one or more embodiments described herein;

2

FIG. 11 illustrates an example selection of particular applications to use with particular SSPs, in accordance with some embodiments;

FIG. 12 illustrates an example of application-anonymous authorization and/or denial of traffic based on a UE with which the traffic is associated, in accordance with some embodiments;

FIG. 13 illustrates an example of application-anonymous authorization and/or denial of traffic based on a UE and a destination with which the traffic is associated, in accordance with some embodiments;

FIGS. 14 and 15 illustrate examples of selectively marking or refraining from providing traffic based on an SSP associated with a given application, in accordance with some embodiments;

FIG. 16 illustrates an example process for selectively applying traffic parameters to traffic associated with a given application based on a UE-side authorization, in accordance with some embodiments;

FIG. 17 illustrates an example process for authorizing and/or rejecting traffic with particular traffic parameters based on a UE from which the traffic was received, in accordance with some embodiments;

FIG. 18 illustrates an example process for authorizing and/or rejecting traffic with particular traffic parameters based on a UE from which the traffic was received as well as authorized parameters for a destination of the traffic, in accordance with some embodiments;

FIG. 19 illustrates an example environment in which one or more embodiments, described herein, may be implemented;

FIG. 20 illustrates an example arrangement of a radio access network (“RAN”), in accordance with some embodiments;

FIG. 21 illustrates an example arrangement of an Open RAN (“O-RAN”) environment in which one or more embodiments, described herein, may be implemented; and

FIG. 22 illustrates example components of one or more devices, in accordance with one or more embodiments described herein.

DETAILED DESCRIPTION OF EXAMPLE EMBODIMENTS

The following detailed description refers to the accompanying drawings. The same reference numbers in different drawings may identify the same or similar elements.

Different applications and/or traffic types, of traffic that is handled via a wireless network, may have different Quality of Service (“QoS”) requirements, routing rules, or the like. Embodiments described herein provide for the determination and enforcement of such QoS requirements, routing rules, etc. for particular traffic types by a wireless network, without requiring exposure of a particular application or traffic type for which such QoS requirements, routing rules, etc. are to be applied. As such, the privacy of users may be preserved, as data pertaining to the usage of particular applications, traffic types, etc. by a given user need not be determined in order for such applications, traffic types, etc. to be handled properly (e.g., to provide QoS treatment, implement routing rules, etc.) by a network, in accordance with embodiments described herein.

As shown in FIG. 1, for example, User Equipment (“UE”) 101, such as a mobile telephone, a tablet device, or the like, may output (at 102) traffic toward application server 103, which may be an application server, a web server, a content streaming system, and/or some other network-accessible

resource. In the example shown here, application server **103** is a particular application server (referred to as “App server_A”), which may be distinct from other application servers **103**. For example, different application servers **103** may have different identifiers, addresses (e.g., Internet Protocol (“IP”) addresses, Uniform Resource Locators (“URLs”), or the like).

In some embodiments, application server **103** may be a “server-side” resource associated with an application executing at UE **101**, such as application **105**. For example, application **105** may execute operations relating to input (e.g., receiving user input such as haptic input, voice input, camera input, etc.) and/or output (e.g., visual output, audible output, haptic output, etc.) associated with a given application (e.g., a gaming application, a content streaming application, a voice call application, etc.) via UE **101**. Application **105** may further communicate (e.g., via network **107**) with application server **103** in order to provide traffic relating to received input to application server **103**, and to receive traffic from application server **103** relating to output to be presented by UE **101**. In the example shown here, application **105** is a particular application (referred to as “App_A”), which may be distinct from other applications **105**. Different applications **105** may communicate with different application servers **103**, and/or multiple different applications **105** may communicate with the same application server **103** (e.g., the same application server **103** may host multiple applications).

Application server **103** may perform computations and/or other operations based on input received from UE **101** (e.g., from application **105**) and/or from one or more other sources. For example, application server **103** may execute “back end” functionality for a gaming application, a video streaming application, an augmented reality (“AR”) application, etc., and may provide traffic to UE **101** (e.g., to application **105**) based on such computations and/or other operations. In this sense, application **105** may be a “client-side” resource of a client-server application executed in conjunction with application server **103**.

The traffic (output at **102**) may include uplink traffic (e.g., traffic destined for application server **103**), a session establishment request (e.g., a protocol data unit (“PDU”) session establishment request, an IP session establishment request, and/or some other sort of session or connection establishment request). In accordance with embodiments described herein, and as indicated in the figure by the strikethrough text, the traffic may not include an identifier of application **105**. For example, omitting the identifier of application **105** may prevent network **107** and/or some other device or system from tracking usage data associated with UE **101**, indicating that UE **101** is sending and/or receiving traffic related to application **105** in particular.

In accordance with embodiments described herein, and as described further below, the traffic (output at **102**) may include an indication that the traffic is destined for application server **103**, and may further include one or more traffic parameters. In some embodiments, the traffic parameters may include a Slicing Service Profile (“SSP”), which may include a requested network slice for the traffic and/or for a session or flow associated with UE **101** and application server **103** (e.g., associated with traffic generated by and/or otherwise associated with application **105**). A “network slice” may refer to a discrete set of resources, devices, systems, Virtualized Network Functions (“VNFs”), etc. associated with network **107**, where different slices provide differentiated levels of service. For example, one particular slice may provide relatively low latency communications

and relatively low throughput, another slice may provide relatively high latency communications and relatively high throughput, yet another slice may provide relatively low latency communications and relatively high throughput, and so on. In some embodiments, the traffic parameters and/or the SSP may include one or more other parameters, such as a requested QoS level, one or more Service Level Agreements (“SLAs”), routing rules, a Data Network Name (“DNN”), and/or other parameters relating to the routing, prioritization, and/or QoS of the traffic.

The parameters included in the traffic (output at **102**) may have been authenticated and/or authorized in, accordance with embodiments described herein, prior to the output (at **102**) of the traffic. For example, as discussed below, application **105** may have already been authorized, by one or more devices or systems associated with network **107** (e.g., by authentication system **109**), for the requested parameters. As further described below, application server **103** (e.g., a “server-side” resource with respect to application **105**) may have also been authorized (e.g., by authentication system **109**) for the requested parameters.

As such, network **107** may determine (at **104**) whether the traffic (output at **102**) is authorized for the requested parameters. For example, authentication system **109** and/or some other device or system associated with network **107** may maintain server-to-SSP information **111**, indicating particular SSPs (or other traffic parameters) that have been authorized for particular resources (e.g., application servers, content provider systems, etc.). As discussed below, server-to-SSP information **111** may be determined or generated as part of an “onboarding” process or other suitable process, in which authentication system **109** authenticates or authorizes particular applications and/or other resources for particular traffic parameters. In this example, authentication system **109** may determine (at **104**) whether application server **103** is authorized for the requested SSP indicated in the traffic (output at **102**) based on server-to-SSP information **111**.

In this manner, since application **105** has previously been authorized for the requested SSP (and this authorization may be maintained and/or verified by an operating system, SIM (“Subscriber Identification Module”), and/or other component of UE **101**), and further since application server **103** has previously been authorized for the requested SSP (and this authorization may be maintained and/or verified by authentication system **109**), the SSP requested in the traffic may be verified by authentication system **109** without requiring that authentication system **109** identify the particular application (e.g., application **105**) with which the traffic is associated. As such, anonymity of application **105** and the privacy of a user of UE **101** may be preserved, while also allowing traffic associated with application **105** to be sent and/or received via network **107** in accordance with suitable traffic parameters.

Assuming that authentication system **109** determines that the traffic (output at **102**) is authorized for the requested parameters (e.g., based on server-to-SSP information **111**), network **107** may allow (at **106**) the traffic to be forwarded, may perform or facilitate the establishment of one or more sessions between UE **101** and application server **103**, may assign the traffic and/or session to a particular flow identifier and/or network slice, and/or may otherwise route the traffic to application server **103** in accordance with the requested traffic parameters.

In situations where authentication system **109** determines that the traffic (output at **102**) is not authorized for the requested parameters, network **107** may still allow (at **108**) the forwarding of the traffic to application server **103**, the

establishment of one or more sessions or flows between UE 101 and application server 103, or the like, but with different parameters than the requested parameters. For example, authentication system 109 may assign a “default” set of parameters or other set of parameters to the traffic. For example, the “default” set of parameters may include a “default” network slice, a “default” QoS level, or the like. Additionally, or alternatively, when determining that the traffic is not authorized for the requested parameters, authentication system 109 may cause (at 110) the traffic to be dropped (e.g., not forwarded to application server 103) and/or may cause a session establishment request to be rejected. In some embodiments, such dropping and/or rejecting (at 110) may include outputting a notification (e.g., a rejection notification, a non-acknowledgement (“NACK”) notification, etc.) to UE 101, indicating that the traffic was not forwarded and/or that a requested session was not established. In this manner, requests for unauthorized traffic parameters may be rejected, thus preserving the efficient operation of network 107.

FIG. 2 illustrates an example of application-anonymous authorization and/or denial of traffic associated with different applications and/or UEs, with particular requested traffic parameters, in accordance with some embodiments. For example, as shown, network 107 may receive various traffic and/or requests (e.g., session establishment requests and/or other requests) associated with different applications. The traffic may include traffic from a first application (e.g., “App_A traffic”), traffic from a second application (e.g., “App_B traffic”), and so on. The traffic may also include unauthorized traffic, which will be discussed below.

The traffic may be received from one particular UE 101, and/or from a group of UEs 101. For the sake of example, assume that each traffic is received from one particular UE 101. As shown, the traffic may include traffic parameters, such as a particular SSP (e.g., a requested SSP, which may include a requested network slice, a requested QoS level, a requested routing policy, etc.) and an indication of a particular destination (e.g., an IP address, a URL, a Fully Qualified Domain Name (“FQDN”), and/or some other indicator of the destination of the traffic and/or an endpoint with which to establish a session, such as a PDU session, an IP session, or the like).

For example, as shown, the App_A traffic may include a first requested SSP (“e.g., SSP_A”) and a first destination (e.g., “App Server_A”), the App_B traffic may include a request for SSP_A and an indication of App Server_A as the destination, the App_C traffic may include a request for SSP_A and indication of App Server_B as the destination, and the App_E traffic may include a request for SSP_B and an indication of App Server_A as the destination. The first unauthorized traffic (e.g., “Unauthorized traffic_A”) may include a request for SSP_A and an indication of App Server_D as the destination, and Unauthorized traffic_B may include a request for SSP_B and an indication of App Server_B as the destination. While shown as being associated with a given application (e.g., App_A traffic associated with an application referred to as “App_A,” App_B traffic associated with an application referred to as “App_B,” and so on), as mentioned above, the traffic may not itself include an actual identifier of the application. Additionally, or alternatively, the traffic may include an encrypted value, a hashed value, a fingerprint, or an otherwise secret or secure value that includes and/or is derived from an application identifier of the respective application with which the traffic is associated. Such value may be “secret” or “secure” inasmuch as network 107 and/or authentication system 109 may not

possess a decryption key or other mechanism by which to ascertain such application identifier.

For the purposes of this example, assume that the traffic does not include an identifier of the respective application with which the traffic is associated. In some embodiments, the received traffic may include one or more data packets, a session establishment request, or other traffic related to (e.g., generated by) particular applications executing on one or more respective UEs 101.

As shown, authentication system 109 may accept and/or reject (at 202) the respective traffic based on comparing attributes or parameters of the traffic to authorization information 211. For example, as discussed above, authentication system 109 may “accept” the traffic by routing the traffic toward its respective destination using the requested parameters (e.g., the requested SSP), facilitating the establishment of one or more respective sessions using the requested parameters, indicating authorization of the traffic to one or more devices or systems of authentication system 109 (e.g., a Session Management Function (“SMF”), a Packet Data Network (“PDN”) Gateway Control function (“PGW-C”), and/or some other device or system), and/or otherwise allowing the traffic to traverse network 107 using the requested parameters associated with the traffic. On the other hand, as discussed above, “rejecting” the traffic may include dropping the traffic and/or allowing the traffic to traverse network 107 using different parameters than the requested parameters (e.g., “default” parameters or other parameters).

As shown, authentication system 109 may maintain authorization information 211, indicating particular SSPs for which particular application servers 103 are associated. For example, authorization information 211 may indicate that App Server_A is authorized for SSP_A and SSP_B, that App Server_B is authorized for SSP_A, that App Server_C is authorized for SSP_A and SSP_B, and that App Server_D is authorized for SSP_B. In some embodiments, authorization information 211 may further include information regarding particular applications associated with particular application servers 103. For example, as shown, authorization information 211 may indicate that App_A, App_B, and App_E are associated with App Server_A, that App_C is associated with App Server_B, and so on. The information indicated by authorization information 211 may be provided and/or determined during an onboarding process, as discussed below, in which authentication system 109 may receive or otherwise determine that particular applications and/or application servers 103 are authorized for particular SSPs.

While authentication system 109 may in some embodiments maintain information associated with particular applications, such information may not be used during the determination (at 202) of whether particular traffic should be accepted and/or rejected. For example, the determination (at 202) may be based on a requested SSP and traffic destination, but not based on an identifier of any particular application. In some embodiments, one or more other factors may be used in the determination (at 202), such as an identifier of a particular UE 101 from which the traffic was received, a geographical location of UE 101, a particular radio access technology (“RAT”) via which UE 101 is connected to a RAN, and/or other factors. In some such embodiments, authentication system 109 may maintain (e.g., in authorization information 211 and/or via some other mechanism) information based on which such determination may be made based on these other factors. For example, in this manner, the authorization of whether particular traffic should be authorized for a requested SSP may further be based on

other factors, where such other factors may be separate from and/or may not include an application identifier.

As discussed below, authentication system 109 may use the information associating particular applications to particular application servers 103 in order to authenticate such applications when these applications are provided (e.g., for installation) to one or more UEs 101. In some embodiments, the information identifying particular applications may include an application identifier, an application fingerprint, a hashed value or some other value derived from the application identifier and/or an installation package associated with the application, a randomly generated value, and/or some other unique identifier.

FIG. 3 illustrates an example signal flow that may be used to securely provide for application-specific traffic treatment without exposing an application identifier to network 107. As shown, for example, application server 103 may output (at 305) an onboarding request to authentication system 109. The request may be output via an application programming interface (“API”), a portal, and/or some other suitable communication pathway. The onboarding request may include an identifier or address associated with application server 103 (e.g., an IP address, a URL, an FQDN, etc.), as well as a particular application or information associated with an application. For example, authentication system 109 may receive an application installation package associated with the application, an executable file or set of files associated with the application, an application identifier, and/or some other information associated with or derived from the application.

In some embodiments, in lieu of an explicit onboarding request from application server 103, authentication system 109 may identify a particular application or set of applications for which particular traffic treatment is to be provided. For example, authentication system 109 may receive an indication from some other device or system of particular applications or application servers for which a particular set of rules, routing policies, QoS levels, etc. are to be applied.

While not described at length here, authentication system 109 may perform any suitable authentication and/or authorization process to determine that the application (indicated at 305) is associated with a particular SSP or other traffic parameters. For example, authentication system 109 may receive or maintain information associating particular applications, types of applications, particular application servers, etc. with particular SSPs or other traffic parameters. Based on such authentication and/or authorization, authentication system 109 may store (at 310) information indicating that application server 103 is authorized for a particular SSP. For example, authentication system 109 may store an association between an address of application server 103 and a particular SSP (e.g., as similarly discussed above with respect to authorization information 211). Authentication system 109 may further generate a token associated with the particular SSP, and/or may select a token from a pool of tokens (e.g., where each token of the pool of tokens is associated with a respective SSP). The token may be and/or may include, for example, an alphanumeric string, a random or pseudorandom value, and/or some other type of suitable information.

In some embodiments, authentication system 109 may maintain information indicating that the token is associated with the application. For example, the application itself (e.g., an application install package, an application executable, etc.) may be unmodified, while authentication system 109 maintains suitable correlation information indicating that the application is associated with the selected or gen-

erated token. Additionally, or alternatively, authentication system 109 may regenerate, recompile, etc. the application (e.g., the application installation package, the application executable file(s), etc.) to include the token and/or information from which the token may be derived.

Further, authentication system 109 may generate a signature that is unique to the particular application and the token. For example, authentication system 109 may perform a cryptographic hashing function and/or some other suitable function on the received application. For example, such function may be performed on an application installation package, an application identifier, and/or other information associated with the application. In some embodiments, the signature may further be based on the token. For example, as noted above, authentication system 109 may, in some embodiments, regenerate, recompile, etc. the application to include the token; in such embodiments, the hashing function or other suitable function performed on the regenerated or recompiled application may therefore be based on the application and the token. Additionally, or alternatively, in embodiments where the application is not regenerated or recompiled, the hashing function or other suitable function may take the application and the token as inputs, and generate the signature based on these separate inputs.

Authentication system 109 may output (at 315) the generated or selected token to application server 103. For example, authentication system 109 may indicate a successful validation of the application, and may provide the token to application server 103. In embodiments where authentication system 109 regenerates, recompiles, etc. (at 310) the application, authentication system 109 may provide the token and the application as a unified application install package, application executable file(s), etc. In embodiments where authentication system 109 does not regenerate, recompile, etc. (at 310) the application, authentication system 109 may provide the token with an application identifier or other suitable information based on which application server 103 may associate (at 320) the application with the separately provided token.

As further shown, authentication system 109 may provide (at 325) a token-to-SSP mapping to one or more UEs, such as a particular UE 101. FIG. 4 illustrates an example data structure 400 that may represent a particular token-to-SSP mapping. For example, as shown, a set of tokens (Token_A, Token_B, Token_C, and Token_D) may each be associated with a particular SSP. For example, Token_A may be associated with SSP_A, Token_B may be associated with SSP_B, Token_C may be associated with SSP_C, and Token_D may be associated with SSP_A. That is, in some embodiments, multiple tokens may be associated with the same SSP. For example, situations may arise where different tokens are generated (e.g., at 310) for the same application, and/or for different applications for which the same SSP is to be applied.

Returning to FIG. 3, authentication system 109 may provide (at 325) the token-to-SSP mapping (e.g., as reflected in data structure 400 and/or a portion thereof) to UE 101 as part of an Over-the-Air (“OTA”) update process, as part of a provisioning process, based on a request from UE 101 (e.g., after UE 101 downloads and/or executes the application, after UE 101 accesses an “app store” or other repository from which the application may be obtained, or the like), and/or at some other time. In some embodiments, authentication system 109 may provide (at 325) the mapping based on selecting and/or generating (at 310) the token for the application, while in some embodiments authentication system 109 may provide the mapping prior to selecting the

token for the application. That is, in some embodiments, the mapping may be “pushed” to UE 101 (e.g., without a request from UE 101, and/or independent of any such requests from UE 101) by authentication system 109 and/or may be “pulled” by UE 101 from authentication system 109. As discussed below, this mapping may be used by UE 101 to request particular SSPs that have been authorized (e.g., at 310) for particular applications. In some embodiments, UE 101 may store the mapping in a SIM (“Subscriber Identification Module”) card, a Universal Integrated Circuit Card (“UICC”), and/or some other secure storage (e.g., where such storage is “secure” inasmuch as the mapping may not be modified and/or accessed by applications installed on or executing at UE 101).

Application server 103 and authentication system 109 may further perform (at 330) a validation process of the application with the token (e.g., as provided at 315). For example, application server 103 may periodically and/or intermittently provide the application and the token (e.g., the application regenerated and/or recompiled with the token, and/or the application with the token separately) for validation by authentication system 109. Additionally, or alternatively, authentication system 109 may obtain the application and token from application server 103 and/or from a repository from which the application may be obtained (e.g., an “app store” and/or some other suitable resource). Authentication system 109 may verify (at 335) that the application and token, as provided by application server 103 and/or other suitable resource (e.g., an “app store”) match the previously authorized (at 310) parameters for the application. In this manner, authentication system 109 may verify that a different application has not been associated (by application server 103) with the authorized SSP, and/or that a different SSP has not been associated with the application. That is, authentication system 109 may verify that the application, available for download to UE 101, is provided with the authorized parameters and not with other parameters.

For example, authentication system 109 may generate (at 335) a signature based on the obtained (at 330) application and token. This signature may be generated in the same manner (e.g., using the same function or functions) as the previously generated (at 310) signature. Authentication system 109 may compare the generated (at 335) signature to the previously generated (at 310) signature. If these signatures match, authentication system 109 may authenticate the application and token, as provided (at 330) by application server 103. If, on the other hand, these signatures do not match, authentication system 109 may revoke an authorization for the particular SSP (identified at 310) for application server 103. For example, authentication system 109 may remove an association between application server 103 (e.g., an address of application server 103) and the SSP. Additionally, or alternatively, authentication system 109 may cause the application to be removed from the resource from which the application was obtained (at 330), may notify application server 103 that the authorization of application server 103 to utilize the SSP has been revoked, may notify application server 103 that the token for the application does not match an authorized token, or perform other suitable operations.

Assuming that the application and token have been validated (at 335), UE 101 may at some point receive (at 340) the application and token from application server 103 and/or some other suitable resource. For example, UE 101 may request the application from an “app store” or other repository, the application may be pushed to UE 101 (e.g., as part

of an OTA update procedure), etc. UE 101 may further identify the token associated with the received application. For example, when installing the application, the token may be extracted from the application installation package and/or may otherwise be identified by UE 101.

Based on the identified token, UE 101 may identify a particular SSP associated with the application. For example, UE 101 may identify a particular SSP associated with the token based on the token-to-SSP mapping received (at 325). As noted above, this mapping may be maintained in a SIM card, a UICC, and/or other suitable storage device associated with UE 101. Additionally, or alternatively, UE 101 may output a request to authentication system 109 for an authorized SSP associated with the token after identifying the token received with the application (at 340).

UE 101 may accordingly store (at 345) information associating the application with the identified SSP. For example, the obtained application may include an application identifier, which may be a name of the application, a random or pseudorandom string, a hash of an application executable file, and/or some other suitable identifier. UE 101 may associate such identifier with the SSP that has been authorized for the application.

At some point, the application may output (at 350) traffic (e.g., destined for application server 103), where such traffic may include and/or may cause a session establishment request (e.g., a PDU session establishment request, an IP session establishment request, etc.), may include application traffic, and/or other outbound traffic. UE 101 may identify the SSP associated with the application, and may output the traffic to application server 103 (e.g., where the application may indicate the address of application server 103), along with an indication or request for the identified SSP to be applied to the traffic.

For example, as shown in FIG. 5, particular application 501 (e.g., referred to as “App_A”) may have been identified (e.g., at 345) as being associated with or including a particular token, and UE 101 may further have identified (e.g., based on data structure 400) that this token is associated with a particular SSP (e.g., SSP_A). When application 501 outputs (at 502) outbound traffic, outputs a request to output outbound traffic, and/or requests a session to be established between UE 101 and application server 103, operating system (“OS”) and/or kernel 503 (and/or some other suitable component of UE 101) may identify whether a particular SSP has been identified for application 501. For example, OS/kernel 503 may query or perform a lookup in a SIM card, UICC, etc. (e.g., UICC 504) based on an application identifier associated with application 501. In this example, UICC 504 may provide (at 504) information indicating that application 501 is associated with a particular SSP (e.g., SSP_A). Thus, prior to outputting the traffic, OS/kernel 503 may modify the traffic to include a request for SSP_A, and may output (at 506) the traffic with the request for SSP_A. For example, OS/kernel 503 may add header information indicating the requested SSP. Additionally, or alternatively, OS/kernel 503 may output a separate request (e.g., to one or more suitable components of network 107, such as to a User Plane Function (“UPF”), a Packet Data Network (“PDN”) gateway (“PGW”), a SMF, a Serving Gateway (“SGW”), authentication system 109, and/or some other suitable device or system) to authorize UE 101 to communicate (e.g., with application server 103) according to the identified SSPA.

Returning to FIG. 3, authentication system 109 may verify (at 355) an association between application server 103 and the requested SSP. As similarly described above, authen-

11

tication system 109 may allow and/or reject (at 360) the traffic with the requested SSP, which may include facilitating or performing a session establishment between application server 103 and UE 101 according to the requested SSP and/or forwarding the outbound traffic to application server 103 according to the requested SSP if authentication system 109 has verified (at 355) that application server 103 is authorized for the requested SSP. On the other hand, as also discussed above, authentication system 109 may drop the traffic, apply a different SSP (e.g., a “default” SSP, a “best effort” SSP, etc., which may be different from the requested SSP), and/or perform other suitable operations when determining (at 355) that application server 103 is not authorized for the requested SSP.

FIG. 6 illustrates an example process 600 for authorizing a particular application and application server 103 for a given set of traffic parameters (e.g., a particular SSP, a particular QoS level, a particular routing policy, etc.). In some embodiments, some or all of process 600 may be performed by authentication system 109. In some embodiments, one or more other devices may perform some or all of process 600 (e.g., in concert with, and/or in lieu of, authentication system 109).

As shown, process 600 may include receiving (at 602) a request, from application server 103, to associate a particular application with a particular set of traffic parameters. For example, authentication system 109 may receive an onboarding request from a particular application server 103 to authorize a particular application for a particular SSP, a particular QoS level, etc. In some embodiments, the request may include an identifier of the application, an application installation package, an application executable file, and/or other suitable information associated with the application. In some embodiments, the onboarding request may specify a particular set of traffic parameters.

Process 600 may further include determining (at 604) that application server 103 and/or the application are authorized for the requested traffic parameters. For example, authentication system 109 may maintain information correlating particular application servers 103 and/or applications to particular sets of traffic parameters. Such information may be received from, for example, a Home Subscriber Server (“HSS”), a Unified Data Management function (“UDM”), or some other source. In some embodiments, the request (at 602) may not include a requested set of traffic parameters, and authentication system 109 may identify the authorized traffic parameters for the application and/or for application server 103 based on information received from an HSS, a UDM, or some other source. In other words, the request (at 602) may include both a query for an authorized set of traffic parameters for a given application server 103 and/or application, as well as an authorization request for such traffic parameters.

Process 600 may additionally include generating and/or selecting (at 606) a token based on the authorized set of traffic parameters. For example, as discussed above, authentication system 109 may generate a token that is unique to the application and/or to application server 103, and/or may select a token from a pool of tokens that are each associated with a particular set of traffic parameters. Thus, the selected and/or generated token may indicate and/or otherwise be associated with the authorized traffic parameters for application server 103 and/or for the application.

In some embodiments, generating and/or selecting (at 606) the token may include maintaining information associating the token with the application. As discussed above, for example, authentication system 109 may store mapping

12

information associating the application (e.g., the application identifier, an application install package, etc.) with the token. Additionally, or alternatively, authentication system 109 may regenerate, recompile, etc. the application (e.g., the application install package, one or more application executable files, etc.) with the token, such that the token is embedded or otherwise included in the application. The association of the token with the application and/or the compiling and/or generating of the application with the token may be referred to herein as a “tokenized application.”

Process 600 may also include generating (at 608) a signature based on the tokenized application. For example, as discussed above, authentication system 109 may perform a hashing function and/or other suitable function on the tokenized application to identify a signature associated with the tokenized application. As discussed above, this signature may be used to verify that an application, as provided by application server 103 to one or more UEs 101, is provided with the authorized token (e.g., as opposed to being provided with another token indicating a non-authorized set of traffic parameters).

Process 600 may further include outputting (at 610) the tokenized application to application server 103. Application server 103 may proceed to make the tokenized application available for download to one or more UEs 101. Additionally, or alternatively, authentication system 109 may provide the tokenized application to a third party system, such as a repository from which the tokenized application may be obtained by one or more UEs 101. Such repository may include an “app store” or other similar resource. When provided by authentication system 109, such tokenized application may be “certified,” where certification indicates that the traffic parameters associated with the tokenized application have been authorized by authentication system 109. In some embodiments, the third party system may be a system to which application server 103 does not have the ability or access to modify the tokenized application, and/or where such access or modification may remove the certification. In some embodiments, when the certification is removed and/or when the tokenized application is otherwise modified or invalidated, authentication system 109 may revoke the authorization for the application and/or application server 103 (e.g., as determined at 604).

FIG. 7 illustrates an example process 700 for obtaining authorization for a particular set of traffic parameters for a particular application. In some embodiments, some or all of process 700 may be performed by application server 103. In some embodiments, one or more other devices may perform some or all of process 700 in concert with, and/or in lieu of, application server 103.

As shown, process 700 may include requesting (at 702) authorization for a particular set of traffic parameters for a particular application. For example, application server 103 may output a request to authentication system 109, indicating a requested set of traffic parameters for a given application hosted by application server 103. Additionally, or alternatively, application server 103 may provide information associated with the application (e.g., an application identifier, an application install package, etc.) without indicating a set of requested traffic parameters. For example, application server 103 may query authentication system 109 for the requested traffic parameters associated with the application.

Process 700 may further include receiving (at 704) a tokenized version of the application, based on the traffic parameters authorized (e.g., by authentication system 109) for the application. For example, as discussed above, the

13

tokenized application may include a modified install package, application executable file, etc. that includes a token that is associated with the authorized traffic parameters. Additionally, or alternatively, as discussed above, the tokenized application may have been provided to a third party system. In such embodiments, application server 103 may receive a link (e.g., a URL, an IP address, and/or some other type of locator) to the tokenized application.

Process 700 may additionally include receiving (at 706) a request for the application from a given UE 101. For example, UE 101 may request to “download” the particular application.

Process 700 may also include requesting (at 708) validation of the tokenized application. For example, as discussed above, authentication system 109 may provide the tokenized application to authentication system 109, which may validate the tokenized application by generating a signature based on the tokenized application and comparing the signature to a previously generated signature for the tokenized application. In this manner, authentication system 109 may verify that the token and/or traffic parameters for the application, to be provided to UE 101, do not deviate from the authorized token and/or traffic parameters. In some embodiments, the validation (at 708) may be performed after and/or based on the request (at 706) for the application. In some embodiments, the validation (at 708) may be performed independently of any such requests.

Process 700 may further include receiving (at 710) an indication of the validation of the tokenized application. For example, authentication system 109 may indicate that the tokenized application is valid (e.g., the token included in or associated with the application matches the authorized token), based on comparing a signature associated with the tokenized application (as included in the request at 708) to a previously generated signature associated with the tokenized application.

Process 700 may additionally include providing (at 712) the validated tokenized application to UE 101. For example, application server 103, an “app store,” and/or some other source may provide the validated tokenized application to UE 101. In some embodiments, the validation (at 710) of the tokenized application may be a requirement and/or precondition of the application being available from application server 103, the “app store,” and/or other suitable resource. Once the tokenized application is received by UE 101, UE 101 may use the token to identify and implement the authorized traffic parameters associated with the application, without indicating the identity of the application to a network via which the traffic is sent.

For example, FIG. 8 illustrates an example process 800 for identifying and implementing the authorized traffic parameters associated with an application, without indicating the identity of the application to a network via which the traffic is sent. In some embodiments, some or all of process 800 may be performed by UE 101. In some embodiments, one or more other devices may perform some or all of process 800 in concert with, and/or in lieu of, UE 101.

As shown, process 800 may include receiving (at 802) token-to-traffic parameter mapping information. For example, UE 101 may receive, from authentication system 109 and/or some other source, information associating particular traffic parameters (e.g., SSPs, QoS levels, or the like) to particular tokens. In some embodiments, such information may be “pushed” to UE 101, such as part of an OTA process and/or some other procedure which may not necessarily be dependent on the installation of any given application. In some embodiments, UE 101 may store the token-to-traffic

14

parameter mapping information in a SIM card, a UICC, and/or some other storage device associated with UE 101. In some embodiments, UE 101 may “protect” such information from modification or access by applications or users, such that the token-to-traffic parameter mapping may not be changed other than by OS/kernel 503 or other suitable component of UE 101.

Process 800 may further include obtaining (at 804) a tokenized application. For example, UE 101 may download a given application (e.g., an install package, an application executable file, etc.) from application server 103, an “app store,” or other suitable resource. In some embodiments, the application may be “pushed” to UE 101 (e.g., as part of an OTA update procedure, a provisioning procedure, and/or some other suitable procedure). From the standpoint of UE 101 and/or a user of UE 101, obtaining the tokenized application may not be a different procedure than obtaining other applications that are not tokenized in the manner described above. For example, UE 101 may obtain two applications from an “app store” using the same or a similar procedure (e.g., clicking an “install” button, a “download” button, etc.), where one application is tokenized (e.g., includes a token in accordance with embodiments described herein, such as an embedded token) while another application is not tokenized (e.g., does not include a token).

Process 800 may additionally include extracting and/or identifying (at 806) a token associated with the application. For example, an application install package may include the token, the installation of the application may include extracting the token, and/or the token may be extracted and/or identified in some other suitable manner.

Process 800 may also include identifying (at 808) traffic parameters for the application based on the extracted and/or identified token. For example, UE 101 may perform a lookup in the token-to-traffic parameter mapping (received at 802) to identify a particular set of traffic parameters associated with the application. Additionally, or alternatively, UE 101 may query authentication system 109 and/or some other suitable device or system with the token, and may receive the particular set of traffic parameters in response to the query. In some embodiments, the query may additionally, or alternatively, include an identifier of the application, based on which authentication system 109 may identify the authorized set of traffic parameters.

Process 800 may further include identifying (at 810) outbound application traffic associated with the application. For example, OS/kernel 503, a modem, and/or some other component of UE 101 that handles the egress of traffic from UE 101 may identify traffic that has been generated and/or output by the application and is to be output from UE 101. For example, the application may make an API call to OS/kernel 503 to output traffic.

Process 800 may additionally include applying (at 812) the traffic parameters to the outbound application traffic. For example, UE 101 (e.g., OS/kernel 503) may add header information to the traffic indicating the traffic parameters and/or may otherwise modify the traffic based on the traffic parameters.

FIG. 9 illustrates an example process 900 for authorizing and/or rejecting traffic with particular traffic parameters based on authorized parameters for a destination of the traffic. In some embodiments, some or all of process 900 may be performed by authentication system 109. In some embodiments, one or more other devices may perform some or all of process 900 in concert with, and/or in lieu of, authentication system 109. For example, as discussed above, an SGW, a UPF, a PGW, a router, and/or some other network

15

component that routes, forwards, and/or otherwise handles traffic may perform some or all of process 900 in concert with authentication system 109.

As shown, process 900 may include receiving (at 902) traffic with particular requested traffic parameters from UE 101. For example, as discussed above, the received traffic may include header information and/or may otherwise indicate a requested SSP, network slice, QoS level, routing policy, and/or other traffic parameters. As further discussed above, such traffic parameters may have been added by UE 101 (e.g., OS/kernel 503) based on a set of traffic parameters identified by UE 101 as being associated with a particular application. The received (at 902) traffic may, in some embodiments, not include an identifier of the particular application. In this manner, the usage of the particular application may be considered as being “anonymous” with respect to authentication system 109.

Process 900 may further include identifying (at 904) a destination associated with the traffic. For example, authentication system 109 may identify an IP address, a URL, an FQDN, and/or other identifier based on header information or other suitable information associated with the traffic.

Process 900 may additionally include determining (at 906) whether the requested traffic parameters are authorized for the destination. For example, as discussed above, authentication system 109 may determine whether application server 103, to which the traffic is destined, has been authorized for the requested traffic parameters (e.g., based on an “onboarding” or other similar process, as discussed above).

If the destination has been authorized for the requested traffic parameters (at 906—YES), then process 900 may also include allowing and/or forwarding (at 908) the traffic with the requested parameters. For example, authentication system 109 may apply a requested SSP to the traffic, forward the traffic via a requested network slice, add header information indicating an authorized QoS level, or the like. Additionally, or alternatively, authentication system 109 may output an indication to another network device (e.g., to an SGW, PGW, UPF, router, and/or other network device) that the traffic has been authorized for the requested traffic parameters, where such network device may in turn apply the traffic parameters.

If, on the other hand, the destination has not been authorized for the requested traffic parameters (at 906—NO), then process 900 may further include rejecting (at 910) the requested traffic parameters. For example, as discussed above, authentication system 109 may drop the traffic, and/or may forward or otherwise process the traffic with a set of parameters that may be different from the requested parameters. For example, authentication system 109 may apply “default” parameters to the traffic. As similarly discussed above, authentication system 109 may provide an indication to one or more network devices that the requested parameters have not been approved, and/or may otherwise forward the traffic without applying the requested traffic parameters.

Embodiments described herein provide for the determination and enforcement of QoS requirements, routing rules, etc. for particular UEs and/or for particular applications executing at UEs. As similarly discussed above, such determination and enforcement may be performed without requiring exposure of a particular application or traffic type for which such QoS requirements, routing rules, etc. are to be applied. In accordance with some embodiments, particular applications may be eligible for particular SSPs (e.g., QoS requirements, routing rules, etc., as discussed above), and the use of a given SSP for a given SSP-eligible application

16

may be enabled or disabled at a UE (e.g., by a user of the UE). Such selective enabling or disabling of an SSP for an application may allow the user of the UE at which the application is executing to have granular control over the use of QoS treatment, routing, etc. of such traffic, which may be useful in situations where such QoS treatment, routing, etc. is limited and/or for which the user is charged extra.

For example, the network may offer an “enhanced gaming service,” an “enhanced file transfer service,” etc., via which particular applications are authorized to receive particular QoS policies denoted by particular SSPs associated with these “enhanced” services. Further, in some embodiments, a given UE may be allotted a limited amount of usage associated with a given “enhanced” service, SSP, set of QoS policies, etc. The limited amount of usage may be, for example, a maximum allotment of traffic sent and/or received via the “enhanced” service and/or some other measure of limited usage associated with a given application.

Situations may arise where multiple applications which are eligible for a particular “enhanced” service or SSP are installed at the UE, but the user of the UE desires not to utilize the “enhanced” service or the SSP for all of these applications, as such usage may consume a limited amount of usage associated with the “enhanced” service or the SSP for the UE. As described herein, the user may be able to select particular applications to use particular SSPs, “enhanced” services, QoS treatment, etc., which may be useful in situations (such as in a limited allotment scenario, as noted above) in which a user may wish to refrain from using such SSPs, “enhanced” services, etc. even when an application is eligible (e.g., authorized by the network) to utilize such SSPs, “enhanced” services, etc.

As shown in FIG. 10, for example, UE 101 may register (at 1002) with authentication system 109 for particular SSPs. Examples herein are described in the context of SSPs, where similar concepts may apply to “enhanced” services, QoS treatment of particular traffic, etc. The registration (at 1002) may be part of, or in conjunction with, a registration (at 1004) of UE 101 with UE information repository 1001, which may be, may include, and/or may be communicatively coupled to one or more network elements that authenticate, authorize, and/or otherwise register UE 101 as eligible to receive service associated with one or more particular SSPs. Such network elements may include an HSS, a UDM, a Policy Control Function (“PCF”), a Policy Charging and Rules Function (“PCRF”), and/or some other device or system that maintains and/or provides information regarding particular SSPs that UE 101 is authorized to access. In some embodiments, UE 101 and UE information repository 1001 may communicate directly or indirectly to register UE 101 for a given SSP.

In some embodiments, the registration and/or authentication of UE 101 (at 1002 and/or 1004, respectively) may include receiving, by authentication system 109, a set of SSP parameters associated with UE 101. The SSP parameters may indicate particular SSPs that UE 101 is authorized to access, and/or parameters associated with such access, such as usage limits associated with particular SSPs. In some embodiments, the SSP parameters may indicate a maximum quantity of applications, associated with UE 101, that are authorized to use the given SSP. Such SSP parameters may be defined by an operator of network 107, an operator of a given application server with which a particular application is associated, and/or in some other suitable manner.

Based on the registration and/or authentication (at 1002 and/or 1004, respectively) of UE 101 with a set of SSPs and

SSP parameters associated with the set of SSPs, Authentication system **109** may maintain UE:SSP parameter information **1003**. In some embodiments, UE:SSP parameter information **1003** may include a mapping of one or more identifiers of UE **101** (e.g., a Subscription Permanent Identifier (“SUPI”), a Globally Unique Temporary Identifier (“GUTI”), an IP address, a Mobile Directory Number (“MDN”), an International Mobile Subscriber Identity (“IMSI”), an International Mobile Station Equipment Identity (“IMEI”), and/or some other suitable identifier) to one or more SSPs. In some embodiments, UE:SSP parameter information **1003** may further include constraints, allotments, usage limits, or the like associated with particular SSPs.

In some embodiments, different UEs **101** may be associated with different SSP parameters for the same SSP. For example, a first UE **101** may be associated with a first usage limit (e.g., a first traffic throughput limit, such as 1 GB) for a given SSP, while a second UE **101** may be associated with a second usage limit (e.g., a second traffic throughput limit, such as 2 GB) for the same SSP. Such different SSP parameters for different UEs **101** may be based on different subscription or service levels associated with UEs **101**, a different amount of historical usage associated with UEs **101** (e.g., the first UE **101** may have consumed more usage associated with the SSP than the second UE **101** in the preceding example, thus leaving the second UE **101** with more available throughput), and/or other factors.

As shown in FIG. **11**, UE **101** may execute, include, etc. SSP client **1101**. In some embodiments, SSP client **1101** may be or may include an application, an API, etc. installed at UE **101**. In some embodiments, SSP client **1101** may be component of OS/kernel **503** associated with UE **101**, and/or may otherwise communicate with OS/kernel **503** of UE **101**. SSP client **1101** may identify (at **1102**) applications **1103-1111** installed at UE **101**. For example, SSP client **1101** may communicate with OS/kernel **503** or some other component of UE **101** to obtain a list of applications (e.g., application names or other identifiers) installed at UE **101**. In some embodiments, some or all of applications **1103-1111**, installed at UE **101**, may implement an API via which such applications **1103-1111** communicate with SSP client **1101**. In some embodiments, only applications that are authorized for one or more SSPs (e.g., tokenized applications, as discussed above) may implement an API via which such applications (e.g., applications **1105**, **1109**, and **1111** in this example) communicate with SSP client **1101**. In some embodiments, SSP client **1101** may otherwise identify one or more SSPs associated with applications **1103-1111** installed at UE **101**, such as in a similar manner as discussed above with respect to FIG. **8**.

In this example, assume that SSP client **1101** identifies (at **1102**) that applications **1105**, **1109**, and **1111** are associated with a particular SSP. For example, as similarly discussed above, SSP client **1101** and/or some other component of UE **101** may identify one or more tokens associated with applications **1105**, **1109**, and **1111**, and may further identify (e.g., based on a token-to-SSP mapping as indicated in data structure **400** or other similar information) an SSP associated with the one or more tokens. In this manner, SSP client **1101** may identify (at **1102**) that applications **1105**, **1109**, and **1111** are SSP-eligible applications (e.g., such applications would potentially be authorized by network **107** for the particular SSP, in a manner similarly described above). SSP client **1101** may also identify SSP parameters associated with the given SSP, which may include a maximum quantity of applications for which UE **101** is authorized to use the SSP.

Based on identifying (at **101**) SSP-eligible applications **1105**, **1109**, and **1111**, SSP client **1101** may present user interface (“UI”) **1113**. For example, SSP client **1101** may cause UE **101** to present a “pop-up” notification, alert, or some other type of UI element, indicating that SSP-eligible applications are installed at UE **101**. SSP client **1101** may, for example, automatically present UI **1113** when a new application (e.g., an SSP-eligible application) is installed at UE **101**, when SSP client **1101** is installed or selected at UE **101**, and/or some other time. In some embodiments, SSP client **1101** may include and/or may be associated with an application executing at UE **101**, that a user of UE **101** may manually select or activate, and which may present UI **1113** upon such selection or activation.

UI **1113** may present, for example, a list of SSP-eligible applications. In this example, assume that applications **1105**, **1109**, and **1111** are associated with (e.g., eligible for) the same SSP. In other examples, UE **1113** may present one or more lists of SSP-eligible applications that are associated with multiple SSPs. For example, UE **1113** may include a first display area that indicates a first set of applications associated with a first set of SSPs, and may include a second display area that indicates a second set of applications associated with a second set of SSPs.

As further shown, UI **1113** may include an indication of one or more constraints, parameters, etc. associated with one or more SSPs. For example, as shown, UI **1113** may include an indication that up to two applications may be selected to use with the particular SSP. This indication may be based on SSP parameters received (e.g., at **1002**) during a registration procedure with authentication system **109** and/or some other device or system.

UE **1113** may further include selectable elements, such as check boxes, radio buttons, menu boxes, etc. that may be selected by a user of UE **101** to select one or more applications to use, enable, etc. with an SSP for which such applications are eligible. In the example shown here, applications **1109** and **1111** may have been selected (e.g., as denoted by the “check” mark in the graphical bubbles proximate to names or identifiers of applications **1109** and **1111** in UI **1113**), while application **1105** may not have been selected (e.g., as denoted by the absence of a “check” mark in the graphical bubble proximate to a name or identifier of application **1105** in UI **1113**). As such, as discussed below, UE **101** (e.g., SSP client **1101** and/or some other component of UE **101**) may utilize a particular SSP for traffic associated with selected applications **1109** and **1111**, and may forgo utilizing the particular SSP for traffic associated with unselected application **1105**.

For example, FIG. **12** illustrates example traffic output by UE **101** in accordance with some embodiments. The traffic may include and/or may otherwise be associated with respective traffic parameters (e.g., SSP indicators), as similarly discussed above. In this example, assume that the traffic illustrated in FIG. **12** refers to traffic associated with one particular UE **101**. In practice, similar concepts may apply for traffic associated with multiple UEs **101**. As shown, authentication system **109** may maintain one or more instances of UE:SSP information **1201**. In some embodiments, UE:SSP parameter information **1003** may be or may include one or more instances of UE:SSP information **1201**. Each instance of UE:SSP information **1201** may include an indication of whether a particular UE **101** is authorized for a given SSP associated with network **107**. For example, for the particular UE **101** with which the traffic illustrated in FIG. **12** is associated, UE:SSP information **1201** may

include an indication that such UE **101** is authorized to access SSP_A and SSP_B, while UE **101** is not authorized to access SSP_C.

The example arrangement of UE:SSP parameter information **1003** and/or instances of UE:SSP information **1201** depicted in FIG. **12** is one possible arrangement, in accordance with one or more embodiments. In some embodiments, such information may be arranged, indicated, and/or maintained differently by authentication system **109**. As similarly discussed above, authentication system **109** may receive UE:SSP parameter information **1003** (e.g., one or more instances of UE:SSP information **1201**) from UE **101**, UE information repository **1001**, and/or some other source during a registration procedure associated with UE **101**, an onboarding procedure associated with UE **101**, a connection of UE **101** to one or more base stations of a RAN associated with network **107**, an authentication procedure associated with UE information repository **1001** and/or some other device or system of network **107**, and/or as part of some other suitable procedure.

In some embodiments, authentication system **109** may receive UE:SSP parameter information **1003** and/or UE:SSP information **1201** in an ongoing process, such that authentication system **109** may maintain up-to-date information as such information changes. For example, a particular UE **101** may consume an allotted amount of network resources (e.g., traffic throughput) associated with a particular SSP, and authentication system **109** may receive information from one or more elements of network **107** (e.g., UE information repository **1001** and/or some other suitable device or system) that such UE **101** is no longer authorized to access the particular SSP. For example, the indication that the example UE **101** is not associated with SSP_C may be based on a revocation of a prior authorization of UE **101** to access SSP_C, and/or UE **101** may not have ever been authorized to access SSP_C.

As shown, authentication system **109** may accept and/or reject (at **1202**) traffic, received from UE **101**, based on UE:SSP parameter information **1003** (e.g., UE:SSP information **1201** associated with UE **101**). For example, authentication system **109** may accept traffic received from UE **101** that is associated with SSP_A, as UE:SSP information **1201** associated with UE **101** indicates (in this example) that UE **101** is authorized to access SSP_A. Referring to the example of FIG. **11**, the traffic marked with SSP_A indicators may include traffic from applications **1109** and **1111**. As similarly discussed above, such traffic may, in some embodiments, not include names and/or identifiers of applications **1109** and/or **1111**, thus maintaining privacy of UE **101** inasmuch as the identities of applications utilized by UE **101** are not directly revealed via application traffic.

As further shown, traffic with no SSP marking, such as traffic associated with application **1105**, may be allowed by authentication system **109** to traverse network **107**. For example, not all traffic output by UE **101** may include SSP markings, such as traffic associated with an application that has not been authorized (e.g., by authentication system **109** and/or some other device or system) to access any SSP, and/or traffic associated with an SSP-eligible application that has not been selected by a user of UE **101** to access any SSP.

As also shown, authentication system **109** may accept traffic from UE **101** that is associated with (e.g., marked with) SSP_B. Such traffic may be associated with another application **1107** of UE **101**. As UE:SSP information **1201** indicates that UE **101** is authorized to access SSP_B, authentication system **109** may accept the traffic from application **1107**, with the SSP_B marking, based on such UE:SSP information **1201**.

As additionally shown, authentication system **109** may reject traffic from UE **101** that is associated with SSP_C. For example, UE **101** may not be authorized for SSP_C, as indicated by UE:SSP information **1201**. In some situations, UE **101** may mark traffic for an application (i.e., application **1103**, in this example) incorrectly due to an error, a miscommunication, or a malicious attempt to obtain enhanced services. In such situations, authentication system **109** may reject such traffic based on UE:SSP information **1201** indicating that UE **101** is not authorized to access SSP_C.

As shown in FIG. **13**, similar concepts may apply to accepting or rejecting traffic on the basis of both UE:SSP information **1201** and authorization information **211**. For example, as noted above, authorization information **211** may indicate particular pairs of SSPs and destinations (e.g., application servers **103** and/or other destinations) for which such SSPs are authorized. In this manner, authentication system **109** may perform a network-side authentication that traffic, received from a particular UE **101** and destined to a particular application server **103** (or other suitable destination) is authorized to access a given SSP.

For example, authentication system **109** may maintain UE:SSP parameter information **1003** (e.g., one or more instances of authorization information **211** associated with one or more UEs **101**) and authorization information **211**, based on which authentication system **109** may accept or reject (at **1302**) traffic, sent by one or more UEs **101** via network **107**. As noted above, such accepting or rejecting may be on the basis of a particular UE **101** from which the traffic was received, a particular SSP associated with (e.g., indicated by) the traffic, and a destination of the traffic.

As shown, for instance, traffic **1301** may be received from a first UE **101** (referred to in this example as "UE_A"). For example, one or more network devices of network **107** may receive traffic **1301** via a RAN of a wireless network and/or some other suitable communication pathway. Authentication system **109** may, in some embodiments, implement, may be implemented by, and/or may be communicatively coupled to such one or more network devices. Traffic **1301** may further include an indicator or request for SSP_A, and/or may otherwise be associated with SSP_A. Additionally, traffic **1301** may include an indicator of a particular destination (i.e., "App Server_A," in this example). Authentication system **109** may verify that the pair of the requested SSP_A and the destination App Server_A by determining, based on authorization information **211**, whether SSP_A and App Server_A are an authorized pair (e.g., App Server_A is authorized to receive traffic associated with SSP_A). Authentication system **109** may further verify, based on UE:SSP parameter information **1003** (e.g., UE:SSP information **1201**) that UE_A is authorized to output traffic associated with SSP_A. In this example, authentication system **109** may verify that App Server_A is authorized to receive traffic associated with SSP_A and that UE_A is authorized to output traffic associated with SSP_A. As such, authentication system **109** may accept traffic **1301**, which is sent from UE_A to App Server_A according to QoS parameters associated with SSP_A.

Similarly, authentication system **109** may accept traffic **1303**, based on determining (e.g., based on UE:SSP parameter information **1003** and authorization information **211**) that UE_B, from which traffic **1303** is sent, is authorized for the requested SSP_A, and that App Server_A is authorized to receive traffic associated with SSP_A. Additionally,

21

authentication system 109 may accept traffic 1305, based on determining (e.g., based on UE:SSP parameter information 1003 and authorization information 211) that UE_A, from which traffic 1305 is sent, is authorized for the requested SSP_A, and that App Server_B is authorized to receive traffic associated with SSP_A. As further shown, authentication system 109 may reject traffic 1307 based on determining (e.g., based on UE:SSP parameter information 1003 and/or authorization information 211) that UE_A is not authorized to access SSP_B (e.g., is not authorized to send traffic associated with SSP_B), and/or that App Server_B is not authorized to receive traffic associated with SSP_B.

In some embodiments, authentication system 109 may use more complex rules or criteria to determine whether to accept or reject (at 1302) traffic based on, for example, UE:SSP parameter information 1003 and authorization information 211. For example, authentication system 109 may accept or reject traffic based on particular combinations of three or more traffic attributes. For example, traffic 1309 may be rejected based on rules, policies, and/or other information received by authentication system 109 (e.g., from UE information repository 1001) indicating that UE_B is not to send traffic, using SSP_A, to App Server_B. Such rules, policies, etc. may override the authorization of UE_B to access SSP_A (e.g., as described above with respect to traffic 1303) and the authorization of App Server_B to access SSP_A (e.g., as described above with respect to traffic 1305).

In addition to or in lieu of network-side enforcement of SSPs associated with particular applications, UEs, and/or destinations (e.g., as shown in FIGS. 12 and 13), some embodiments may perform UE-side enforcement or utilization of particular SSPs for particular applications. For example, as shown in FIG. 14, OS/kernel 503 of UE 101 may communicate with SSP client 1101, in order to receive SSP information for particular applications, and/or to authenticate or verify whether a particular application is authorized to utilize a given SSP. In some embodiments, SSP client 1101 and OS/kernel 503 may communicate via an API or other suitable communication pathway. In some embodiments, OS/kernel 503 may implement some or all of the functionality of SSP client 1101.

As similarly discussed above with respect to FIG. 5, a particular application 1105 may have been identified (e.g., by OS/kernel 503, SSP client 1101, and/or some other component of UE 101) as being associated with or including a particular token, and UE 101 may further have identified (e.g., based on data structure 400) that this token is associated with a particular SSP (e.g., SSP_A). UE 101 (e.g., OS/kernel 503, SSP client 1101, and/or some other component of UE 101) may have also identified that application 1105 has been authorized (e.g., via a selection in UI 1113 or some other user selection via UE 101) to utilize SSP_A. For example, as discussed above, such selection or authorization may have been performed as part of a registration process (e.g., at 1002) between UE 101 and authentication system 109 and/or some other device or system.

When application 1105 outputs (at 1402) outbound traffic, outputs a request to output outbound traffic, and/or requests a session to be established between UE 101 and application server 103, OS/kernel 503 (and/or some other suitable component of UE 101) may identify whether a particular SSP has been identified for application 1105 (e.g., whether application is SSP-eligible for SSP_A). For example, OS/kernel 503 may query or perform a lookup in a SIM card, UICC, etc. (e.g., UICC 504) based on an application identifier associated with application 1105. In this example, UICC 504 may provide (at 1404) information indicating that

22

application 1105 is associated with a particular SSP (i.e., is SSP-eligible for SSP_A, in this example).

OS/kernel 503 may further identify whether application 1105 has received UE-level authorization to utilize SSP_A, which may have been received via UI 1113 or some other indication by a user of UE 101 that application 1105 is authorized to utilize SSP_A. For example, OS/kernel 503 may query or otherwise communicate with SSP client 1101 to identify whether application 1105 is authorized (e.g., has been authorized by a user of UE 101) to utilize the SSP for which application 1105 is eligible (i.e., SSP_A, in this example). In this example, SSP client 1101 may indicate (at 1406) that application 1105 is authorized to utilize SSP_A. In this manner OS/kernel 503 may verify that application 1105 has been authorized by both a network to utilize SSP_A (e.g., based on information received at 1404), and by a user of UE 101 to utilize SSP_A (e.g., based on information received at 1406). UE 101 may proceed to output (at 1408) the outbound traffic associated with application 1105, including an indicator or request for SSP_A, based on determining that application 1105 has been authorized by both the network and the user of UE 101 to utilize SSP_A.

In the example of FIG. 15, OS/kernel 503 may receive (at 1502) an indication that application 1107 is SSP-eligible for SSP_A, and may receive (at 1504) an indication that application 1107 has not been authorized (e.g., by a user of UE 101) to utilize SSP_A. For example, the user of UE 101 may have selected different applications of UE 101 to utilize SSP_A (e.g., other than application 1107), or may have selected no applications of UE 101 to utilize SSP_A. Based on determining that application 1107 has not received UE-level authorization to use SSP_A (e.g., based on information received at 1504), OS/kernel 503 may determine that outbound traffic associated with application 1107 should not be output (at 1506) with a request or indicator of SSP_A, even though SSP_A has been indicated as being SSP-eligible for SSP_A (e.g., has received network-side authorization for SSP_A).

FIG. 16 illustrates an example process 1600 for selectively applying traffic parameters to traffic associated with a given application based on a UE-side authorization, in accordance with some embodiments. In some embodiments, some or all of process 1600 may be performed by UE 101 (e.g., by SSP client 1101). In some embodiments, one or more other devices may perform some or all of process 1600 in concert with, and/or in lieu of, UE 101, and/or by one or more other components of UE 101.

As shown, process 1600 may include identifying (at 1602) a token associated with a tokenized application. For example, as similarly discussed above, UE 101 may be associated with one or more applications that have been authorized for a particular set of traffic parameters (e.g., one or more SSPs) by a network, and the tokenized application may be used to securely indicate which applications are associated with which sets of traffic parameters.

Process 1600 may further include identifying (at 1604) traffic parameters for the particular application based on extracting and/or otherwise identifying the token. For example, as similarly discussed above, UE 101 may identify a token-to-traffic parameter mapping in order to identify particular traffic parameters (e.g., an authorized SSP) for the application.

Process 1600 may additionally include requesting (at 1606) device-level (e.g., UE-level) authorization for the application to utilize the identified traffic parameters. For example, SSP client 1101 may present UI 1113, indicating that the particular application is eligible (e.g., authorized by

the network) to utilize the identified traffic parameters (e.g., one or more particular SSPs). UE 101 may receive a selection (e.g., a user selection) via UI 1113, indicating that the traffic parameters are authorized (e.g., by the user) to utilize the identified traffic parameters (e.g., the one or more SSPs). Additionally, or alternatively, SSP client 1101 may receive an indication (e.g., via an API) from one or more other sources, indicating that the particular application has received UE-level authorization to access the identified traffic parameters.

Process 1600 may also include identifying (at 1608) outbound application traffic associated with the particular application. For example, SSP client 1101, OS/kernel 503, and/or some other component of UE 101 may receive outbound traffic associated with the particular application, and/or an indication regarding such outbound traffic associated with the particular application.

Process 1600 may further include selectively applying (at 1610) the traffic parameters for outbound traffic associated with the particular application based on the UE-level authorization. For example, SSP client 1101, OS/kernel 503, and/or some other component of UE 101 may apply the traffic parameters when UE-level authorization to utilize such traffic parameters has been received (e.g., at 1606). On the other hand, SSP client 1101, OS/kernel 503, and/or some other component of UE 101 may refrain from applying the traffic parameters when UE-level authorization to utilize such traffic parameters has not been received. As discussed above, applying the traffic parameters may include marking the traffic with an indicator associated with the traffic parameters (e.g., an indicator of a particular SSP). In some embodiments, such marking may be included in header information of the traffic. As further discussed above, one or more elements of network 107 may authenticate the use of the particular traffic parameters for the traffic based on a network-side authorization procedure, and may accept or reject the traffic with the indicated traffic parameters.

FIG. 17 illustrates an example process for authorizing and/or rejecting traffic with particular traffic parameters based on a particular UE 101 from which the traffic was received, in accordance with some embodiments. In some embodiments, some or all of process 1700 may be performed by authentication system 109. In some embodiments, one or more other devices may perform some or all of process 1700 in concert with, and/or in lieu of, authentication system 109.

As shown, process 1700 may include receiving (at 1702) traffic with particular requested traffic parameters from UE 101. For example, as discussed above, the received traffic may include header information and/or may otherwise indicate a requested SSP, network slice, QoS level, routing policy, and/or other traffic parameters. As further discussed above, such traffic parameters may have been added by UE 101 (e.g., SSP client 1101, OS/kernel 503, and/or some other component of UE 101) based on a set of traffic parameters identified by UE 101 as being associated with a particular application, and further based on receiving UE-level authorization to utilize the traffic parameters. The received (at 1702) traffic may, in some embodiments, not include an identifier of the particular application. In this manner, as discussed above, the usage of the particular application may be considered as being “anonymous” with respect to authentication system 109.

Process 1700 may further include identifying (at 1704) a particular UE 101 associated with the traffic. For example, authentication system 109 may identify an IP address, a SUPI, a GUTI, an IMSI, an IMEI, and/or other identifier of

UE 101 based on header information or other suitable information associated with the traffic.

Process 1700 may additionally include determining (at 1706) whether the requested traffic parameters are authorized for UE 101. For example, as discussed above, authentication system 109 may determine whether UE 101, from which the traffic is received, has been authorized for the requested traffic parameters (e.g., based on a registration procedure or other suitable procedure, as discussed above).

If UE 101 has been authorized for the requested traffic parameters (at 1706—YES), then process 1700 may also include allowing and/or forwarding (at 1708) the traffic with the requested parameters. For example, authentication system 109 may apply a requested SSP to the traffic, forward the traffic via a requested network slice, add header information indicating an authorized QoS level, or the like. Additionally, or alternatively, authentication system 109 may output an indication to another network device (e.g., to an SGW, PGW, UPF, router, and/or other network device) that the traffic has been authorized for the requested traffic parameters, where such network device may in turn apply the traffic parameters.

If, on the other hand, UE 101 has not been authorized for the requested traffic parameters (at 1706—NO), then process 1700 may further include rejecting (at 1710) the requested traffic parameters. For example, as discussed above, authentication system 109 may drop the traffic, and/or may forward or otherwise process the traffic with a set of parameters that may be different from the requested parameters. For example, authentication system 109 may apply “default” parameters to the traffic. As similarly discussed above, authentication system 109 may provide an indication to one or more network devices that the requested parameters have not been approved, and/or may otherwise forward the traffic without applying the requested traffic parameters.

FIG. 18 illustrates an example process for authorizing and/or rejecting traffic with particular traffic parameters based on a particular UE 101 from which the traffic was received, as well as authorized parameters for a destination of the traffic, in accordance with some embodiments. In some embodiments, some or all of process 1800 may be performed by authentication system 109. In some embodiments, one or more other devices may perform some or all of process 1800 in concert with, and/or in lieu of, authentication system 109.

As shown, process 1800 may include receiving (at 1802) traffic with particular requested traffic parameters from UE 101. For example, as similarly discussed above, the received traffic may include header information and/or may otherwise indicate a requested SSP, network slice, QoS level, routing policy, and/or other traffic parameters.

Process 1800 may further include identifying (at 1804) a particular UE 101 and destination associated with the traffic. For example, authentication system 109 may identify an IP address, a SUPI, a GUTI, an IMSI, an IMEI, and/or other identifier of UE 101 based on header information or other suitable information associated with the traffic. Further, authentication system 109 may identify an IP address, a URL, an FQDN, and/or other identifier of the destination based on header information or other suitable information associated with the traffic.

Process 1800 may additionally include determining (at 1806) whether the requested traffic parameters are authorized for UE 101. For example, as discussed above, authentication system 109 may determine whether UE 101, from which the traffic is received, has been authorized for the

requested traffic parameters (e.g., based on a registration procedure or other suitable procedure, as discussed above).

Process **1800** may additionally include determining (at **1808**) whether the requested traffic parameters are authorized for the destination. For example, as discussed above, authentication system **109** may determine whether application server **103**, to which the traffic is destined, has been authorized for the requested traffic parameters (e.g., based on an “onboarding” procedure or other similar procedure, as discussed above).

If UE **101** and the destination have been authorized for the requested traffic parameters (at **1806**—YES and **1808**—YES), then process **1800** may include allowing and/or forwarding (at **1810**) the traffic with the requested parameters. For example, authentication system **109** may apply a requested SSP to the traffic, forward the traffic via a requested network slice, add header information indicating an authorized QoS level, or the like. Additionally, or alternatively, authentication system **109** may output an indication to another network device (e.g., to an SGW, PGW, UPF, router, and/or other network device) that the traffic has been authorized for the requested traffic parameters, where such network device may in turn apply the traffic parameters.

If, on the other hand, UE **101** has not been authorized for the requested traffic parameters (at **1806**—NO) and/or if the destination has not been authorized for the requested traffic parameters (**1808**—NO), then process **1800** may further include rejecting (at **1812**) the requested traffic parameters. For example, as discussed above, authentication system **109** may drop the traffic, and/or may forward or otherwise process the traffic with a set of parameters that may be different from the requested parameters. For example, authentication system **109** may apply “default” parameters to the traffic. As similarly discussed above, authentication system **109** may provide an indication to one or more network devices that the requested parameters have not been approved, and/or may otherwise forward the traffic without applying the requested traffic parameters.

FIG. **19** illustrates an example environment **1900**, in which one or more embodiments may be implemented. In some embodiments, environment **1900** may correspond to a Fifth Generation (“5G”) network, and/or may include elements of a 5G network. In some embodiments, environment **1900** may correspond to a 5G Non-Standalone (“NSA”) architecture, in which a 5G RAT may be used in conjunction with one or more other RATs (e.g., a Long-Term Evolution (“LTE”) RAT), and/or in which elements of a 5G core network may be implemented by, may be communicatively coupled with, and/or may include elements of another type of core network (e.g., an evolved packet core (“EPC”). As shown, environment **1900** may include UE **101**, RAN **1910** (which may include one or more Next Generation Node Bs (“gNBs”) **1911**), RAN **1912** (which may include one or more evolved Node Bs (“eNBs”) **1913**), and various network functions such as Access and Mobility Management Function (“AMF”) **1915**, Mobility Management Entity (“MME”) **1916**, SGW **1917**, Session Management Function SMF/PGW-Control plane function (“PGW-C”) **1920**, Policy Control Function PCF/PCRF **1925**, Application Function (“AF”) **1930**, UPF/PGW-User plane function (“PGW-U”) **1935**, HSS/UDM **1940**, and Authentication Server Function (“AUSF”) **1945**. Environment **1900** may also include one or more networks, such as Data Network (“DN”) **1950**. Environment **1900** may include one or more additional devices or systems communicatively coupled to one or more networks (e.g., DN **1950**), such as

application server **103** and/or authentication system **109**, which may perform one or more operations described above.

The example shown in FIG. **19** illustrates one instance of each network component or function (e.g., one instance of SMF/PGW-C **1920**, PCF/PCRF **1925**, UPF/PGW-U **1935**, HSS/UDM **1940**, and/or **1945**). In practice, environment **1900** may include multiple instances of such components or functions. For example, in some embodiments, environment **1900** may include multiple “slices” of a core network, where each slice includes a discrete set of network functions (e.g., one slice may include a first instance of SMF/PGW-C **1920**, PCF/PCRF **1925**, UPF/PGW-U **1935**, HSS/UDM **1940**, and/or **1945**, while another slice may include a second instance of SMF/PGW-C **1920**, PCF/PCRF **1925**, UPF/PGW-U **1935**, HSS/UDM **1940**, and/or **1945**). The different slices may provide differentiated levels of service, such as service in accordance with different QoS parameters.

The quantity of devices and/or networks, illustrated in FIG. **19**, is provided for explanatory purposes only. In practice, environment **1900** may include additional devices and/or networks, fewer devices and/or networks, different devices and/or networks, or differently arranged devices and/or networks than illustrated in FIG. **19**. For example, while not shown, environment **1900** may include devices that facilitate or enable communication between various components shown in environment **1900**, such as routers, modems, gateways, switches, hubs, etc. Alternatively, or additionally, one or more of the devices of environment **1900** may perform one or more network functions described as being performed by another one or more of the devices of environment **1900**. Devices of environment **1900** may interconnect with each other and/or other devices via wired connections, wireless connections, or a combination of wired and wireless connections. In some implementations, one or more devices of environment **1900** may be physically integrated in, and/or may be physically attached to, one or more other devices of environment **1900**.

UE **101** may include a computation and communication device, such as a wireless mobile communication device that is capable of communicating with RAN **1910**, RAN **1912**, and/or DN **1950**. UE **101** may be, or may include, a radiotelephone, a personal communications system (“PCS”) terminal (e.g., a device that combines a cellular radiotelephone with data processing and data communications capabilities), a personal digital assistant (“PDA”) (e.g., a device that may include a radiotelephone, a pager, Internet/intranet access, etc.), a smart phone, a laptop computer, a tablet computer, a camera, a personal gaming system, an IoT device (e.g., a sensor, a smart home appliance, or the like), a wearable device, an Internet of Things (“IoT”) device, a Machine-to-Machine (“M2M”) device, or another type of mobile computation and communication device. UE **101** may send traffic to and/or receive traffic (e.g., user plane traffic) from DN **1950** via RAN **1910**, RAN **1912**, and/or UPF/PGW-U **1935**.

RAN **1910** may be, or may include, a 5G RAN that includes one or more base stations (e.g., one or more gNBs **1911**), via which UE **101** may communicate with one or more other elements of environment **1900**. UE **101** may communicate with RAN **1910** via an air interface (e.g., as provided by gNB **1911**). For instance, RAN **1910** may receive traffic (e.g., voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE **101** via the air interface, and may communicate the traffic to UPF/PGW-U **1935**, and/or one or more other devices or networks. Similarly, RAN **1910** may receive traffic intended for UE **101** (e.g., from UPF/PGW-U **1935**, AMF **1915**, and/or one or

more other devices or networks) and may communicate the traffic to UE 101 via the air interface.

RAN 1912 may be, or may include, a LTE RAN that includes one or more base stations (e.g., one or more eNBs 1913), via which UE 101 may communicate with one or more other elements of environment 1900. UE 101 may communicate with RAN 1912 via an air interface (e.g., as provided by eNB 1913). For instance, RAN 1910 may receive traffic (e.g., voice call traffic, data traffic, messaging traffic, signaling traffic, etc.) from UE 101 via the air interface, and may communicate the traffic to UPF/PGW-U 1935, and/or one or more other devices or networks. Similarly, RAN 1910 may receive traffic intended for UE 101 (e.g., from UPF/PGW-U 1935, SGW 1917, and/or one or more other devices or networks) and may communicate the traffic to UE 101 via the air interface.

AMF 1915 may include one or more devices, systems, VNFs, etc., that perform operations to register UE 101 with the 5G network, to establish bearer channels associated with a session with UE 101, to hand off UE 101 from the 5G network to another network, to hand off UE 101 from the other network to the 5G network, manage mobility of UE 101 between RANs 1910 and/or gNBs 1911, and/or to perform other operations. In some embodiments, the 5G network may include multiple AMFs 1915, which communicate with each other via the N14 interface (denoted in FIG. 19 by the line marked "N14" originating and terminating at AMF 1915).

MME 1916 may include one or more devices, systems, VNFs, etc., that perform operations to register UE 101 with the EPC, to establish bearer channels associated with a session with UE 101, to hand off UE 101 from the EPC to another network, to hand off UE 101 from another network to the EPC, manage mobility of UE 101 between RANs 1912 and/or eNBs 1913, and/or to perform other operations.

SGW 1917 may include one or more devices, systems, VNFs, etc., that aggregate traffic received from one or more eNBs 1913 and send the aggregated traffic to an external network or device via UPF/PGW-U 1935. Additionally, SGW 1917 may aggregate traffic received from one or more UPF/PGW-U 1935 and may send the aggregated traffic to one or more eNBs 1913. SGW 1917 may operate as an anchor for the user plane during inter-eNB handovers and as an anchor for mobility between different telecommunication networks or RANs (e.g., RANs 1910 and 1912).

SMF/PGW-C 1920 may include one or more devices, systems, VNFs, etc., that gather, process, store, and/or provide information in a manner described herein. SMF/PGW-C 1920 may, for example, facilitate in the establishment of communication sessions on behalf of UE 101. In some embodiments, the establishment of communications sessions may be performed in accordance with one or more policies provided by PCF/PCRF 1925.

PCF/PCRF 1925 may include one or more devices, systems, VNFs, etc., that aggregate information to and from the 5G network and/or other sources. PCF/PCRF 1925 may receive information regarding policies and/or subscriptions from one or more sources, such as subscriber databases and/or from one or more users (such as, for example, an administrator associated with PCF/PCRF 1925).

AF 1930 may include one or more devices, systems, VNFs, etc., that receive, store, and/or provide information that may be used in determining parameters (e.g., quality of service parameters, charging parameters, or the like) for certain applications.

UPF/PGW-U 1935 may include one or more devices, systems, VNFs, etc., that receive, store, and/or provide data

(e.g., user plane data). For example, UPF/PGW-U 1935 may receive user plane data (e.g., voice call traffic, data traffic, etc.), destined for UE 101, from DN 1950, and may forward the user plane data toward UE 101 (e.g., via RAN 1910, SMF/PGW-C 1920, and/or one or more other devices). In some embodiments, multiple UPFs 1935 may be deployed (e.g., in different geographical locations), and the delivery of content to UE 101 may be coordinated via the N9 interface (e.g., as denoted in FIG. 19 by the line marked "N9" originating and terminating at UPF/PGW-U 1935). Similarly, UPF/PGW-U 1935 may receive traffic from UE 101 (e.g., via RAN 1910, SMF/PGW-C 1920, and/or one or more other devices), and may forward the traffic toward DN 1950. In some embodiments, UPF/PGW-U 1935 may communicate (e.g., via the N4 interface) with SMF/PGW-C 1920, regarding user plane data processed by UPF/PGW-U 1935.

HSS/UDM 1940 and AUSF 1945 may include one or more devices, systems, VNFs, etc., that manage, update, and/or store, in one or more memory devices associated with AUSF 1945 and/or HSS/UDM 1940, profile information associated with a subscriber. AUSF 1945 and/or HSS/UDM 1940 may perform authentication, authorization, and/or accounting operations associated with the subscriber and/or a communication session with UE 101.

DN 1950 may include one or more wired and/or wireless networks. For example, DN 1950 may include an Internet Protocol IP-based PDN, a wide area network ("WAN") such as the Internet, a private enterprise network, and/or one or more other networks. UE 101 may communicate, through DN 1950, with data servers, other UEs 101, and/or to other servers or applications that are coupled to DN 1950. DN 1950 may be connected to one or more other networks, such as a public switched telephone network ("PSTN"), a public land mobile network ("PLMN"), and/or another network. DN 1950 may be connected to one or more devices, such as content providers, applications, web servers, and/or other devices, with which UE 101 may communicate.

Application server 103 may include one or more devices, systems, VNFs, or the like, which perform one or more operations described herein. Although referred to as an "application server," in practice, the same or similar operations described with respect to application server 103 may be performed by some other type of device or system, such as a content provider system, a cloud computing system, or the like. Application server 103 may host a server side of one or more applications, where a client side of such applications may be provided to and/or executed by one or more UEs 101. For example, application server 103 may receive traffic from UE 101, perform operations based on the traffic, and output further traffic to UE 101 based on the operations.

Authentication system 109 may include one or more devices, systems, VNFs, or the like, which perform one or more operations described herein. For example, authentication system 109 may authorize particular applications, UEs 101, and/or application servers 103 for particular traffic parameters, such as network slicing parameters, QoS parameters, or the like. Authentication system 109 may maintain information correlating particular application servers 103 and/or UEs 101 to particular sets of parameters, and may provide tokenized versions of appropriate applications to application server 103 and/or other device or system for distribution to one or more UEs 101. Authentication system 109 may further determine whether traffic, output by UEs 101 with particular traffic parameters, is authorized for the traffic parameters indicated in such traffic. Authentication system 109 may communicate with one or more other devices, systems, or VNFs to indicate the authorization or

rejection of certain traffic parameters with respect to traffic from one or more UEs **101**, where such other devices, systems, or VNFs may implement such parameters.

FIG. **20** illustrates an example Distributed Unit (“DU”) network **2000**, which may be included in and/or implemented by one or more RANs (e.g., RAN **1910**, RAN **1912**, or some other RAN). In some embodiments, a particular RAN may include one DU network **2000**. In some embodiments, a particular RAN may include multiple DU networks **2000**. In some embodiments, DU network **2000** may correspond to a particular gNB **1911** of a 5G RAN (e.g., RAN **1910**). In some embodiments, DU network **2000** may correspond to multiple gNBs **1911**. In some embodiments, DU network **2000** may correspond to one or more other types of base stations of one or more other types of RANs. As shown, DU network **2000** may include Central Unit (“CU”) **2005**, one or more Distributed Units (“DUs”) **2003-1** through **2003-N** (referred to individually as “DU **2003**,” or collectively as “DUs **2003**”), and one or more Radio Units (“RUs”) **2001-1** through **2001-M** (referred to individually as “RU **2001**,” or collectively as “RUs **2001**”).

CU **2005** may communicate with a core of a wireless network (e.g., may communicate with one or more of the devices or systems described above with respect to FIG. **19**, such as AMF **1915** and/or UPF/PGW-U **1935**). In the uplink direction (e.g., for traffic from UEs **101** to a core network), CU **2005** may aggregate traffic from DUs **2003**, and forward the aggregated traffic to the core network. In some embodiments, CU **2005** may receive traffic according to a given protocol (e.g., Radio Link Control (“RLC”)) from DUs **2003**, and may perform higher-layer processing (e.g., may aggregate/process RLC packets and generate Packet Data Convergence Protocol (“PDCP”) packets based on the RLC packets) on the traffic received from DUs **2003**.

In accordance with some embodiments, CU **2005** may receive downlink traffic (e.g., traffic from the core network) for a particular UE **101**, and may determine which DU(s) **2003** should receive the downlink traffic. DU **2003** may include one or more devices that transmit traffic between a core network (e.g., via CU **2005**) and UE **101** (e.g., via a respective RU **2001**). DU **2003** may, for example, receive traffic from RU **2001** at a first layer (e.g., physical (“PHY”) layer traffic, or lower PHY layer traffic), and may process/aggregate the traffic to a second layer (e.g., upper PHY and/or RLC). DU **2003** may receive traffic from CU **2005** at the second layer, may process the traffic to the first layer, and provide the processed traffic to a respective RU **2001** for transmission to UE **101**.

RU **2001** may include hardware circuitry (e.g., one or more RF transceivers, antennas, radios, and/or other suitable hardware) to communicate wirelessly (e.g., via an RF interface) with one or more UEs **101**, one or more other DUs **2003** (e.g., via RUs **2001** associated with DUs **2003**), and/or any other suitable type of device. In the uplink direction, RU **2001** may receive traffic from UE **101** and/or another DU **2003** via the RF interface and may provide the traffic to DU **2003**. In the downlink direction, RU **2001** may receive traffic from DU **2003**, and may provide the traffic to UE **101** and/or another DU **2003**.

RUs **2001** may, in some embodiments, be communicatively coupled to one or more Multi-Access/Mobile Edge Computing (“MEC”) devices, referred to sometimes herein simply as “MECs” **2007**. For example, RU **2001-1** may be communicatively coupled to MEC **2007-1**, RU **2001-M** may be communicatively coupled to MEC **2007-M**, DU **2003-1** may be communicatively coupled to MEC **2007-2**, DU **2003-N** may be communicatively coupled to MEC **2007-N**,

CU **2005** may be communicatively coupled to MEC **2007-3**, and so on. MECs **2007** may include hardware resources (e.g., configurable or provisionable hardware resources) that may be configured to provide services and/or otherwise process traffic to and/or from UE **101**, via a respective RU **2001**.

For example, RU **2001-1** may route some traffic, from UE **101**, to MEC **2007-1** instead of to a core network (e.g., via DU **2003** and CU **2005**). MEC **2007-1** may process the traffic, perform one or more computations based on the received traffic, and may provide traffic to UE **101** via RU **2001-1**. In this manner, ultra-low latency services may be provided to UE **101**, as traffic does not need to traverse DU **2003**, CU **2005**, and an intervening backhaul network between DU network **2000** and the core network. In some embodiments, MEC **2007** may include, and/or may implement, some or all of the functionality described above with respect to authentication system **109**.

FIG. **21** illustrates an example O-RAN environment **2100**, which may correspond to RAN **1910**, RAN **1912**, and/or DU network **2000**. For example, RAN **1910**, RAN **1912**, and/or DU network **2000** may include one or more instances of O-RAN environment **2100**, and/or one or more instances of O-RAN environment **2100** may implement RAN **1910**, RAN **1912**, DU network **2000**, and/or some portion thereof. As shown, O-RAN environment **2100** may include Non-Real Time Radio Intelligent Controller (“RIC”) **2101**, Near-Real Time RIC **2103**, O-eNB **2105**, O-CU-Control Plane (“O-CU-CP”) **2107**, O-CU-User Plane (“O-CU-UP”) **2109**, O-DU **2111**, O-RU **2113**, and O-Cloud **2115**. In some embodiments, O-RAN environment **2100** may include additional, fewer, different, and/or differently arranged components.

In some embodiments, some or all of the elements of O-RAN environment **2100** may be implemented by one or more configurable or provisionable resources, such as virtual machines, cloud computing systems, physical servers, and/or other types of configurable or provisionable resources. In some embodiments, some or all of O-RAN environment **2100** may be implemented by, and/or communicatively coupled to, one or more MECs **2007**.

Non-Real Time RIC **2101** and Near-Real Time RIC **2103** may receive performance information (and/or other types of information) from one or more sources, and may configure other elements of O-RAN environment **2100** based on such performance or other information. For example, Near-Real Time RIC **2103** may receive performance information, via one or more E2 interfaces, from O-eNB **2105**, O-CU-CP **2107**, and/or O-CU-UP **2109**, and may modify parameters associated with O-eNB **2105**, O-CU-CP **2107**, and/or O-CU-UP **2109** based on such performance information. Similarly, Non-Real Time RIC **2101** may receive performance information associated with O-eNB **2105**, O-CU-CP **2107**, O-CU-UP **2109**, and/or one or more other elements of O-RAN environment **2100** and may utilize machine learning and/or other higher level computing or processing to determine modifications to the configuration of O-eNB **2105**, O-CU-CP **2107**, O-CU-UP **2109**, and/or other elements of O-RAN environment **2100**. In some embodiments, Non-Real Time RIC **2101** may generate machine learning models based on performance information associated with O-RAN environment **2100** or other sources, and may provide such models to Near-Real Time RIC **2103** for implementation.

O-eNB **2105** may perform functions similar to those described above with respect to eNB **1013**. For example, O-eNB **2105** may facilitate wireless communications between UE **101** and a core network. O-CU-CP **2107** may

perform control plane signaling to coordinate the aggregation and/or distribution of traffic via one or more DUs **2003**, which may include and/or be implemented by one or more O-DUs **2111**, and O-CU-UP **2109** may perform the aggregation and/or distribution of traffic via such DUs **2003** (e.g., O-DUs **2111**). O-DU **2111** may be communicatively coupled to one or more RUs **2001**, which may include and/or may be implemented by one or more O-RUs **2113**. In some embodiments, O-Cloud **2115** may include or be implemented by one or more MECs **2007**, which may provide services, and may be communicatively coupled, to O-CU-CP **2107**, O-CU-UP **2109**, O-DU **2111**, and/or O-RU **2113** (e.g., via an O1 and/or O2 interface).

FIG. **22** illustrates example components of device **2200**. One or more of the devices described above may include one or more devices **2200**. Device **2200** may include bus **2210**, processor **2220**, memory **2230**, input component **2240**, output component **2250**, and communication interface **2260**. In another implementation, device **2200** may include additional, fewer, different, or differently arranged components.

Bus **2210** may include one or more communication paths that permit communication among the components of device **2200**. Processor **2220** may include a processor, microprocessor, or processing logic that may interpret and execute instructions. Memory **2230** may include any type of dynamic storage device that may store information and instructions for execution by processor **2220**, and/or any type of non-volatile storage device that may store information for use by processor **2220**.

Input component **2240** may include a mechanism that permits an operator to input information to device **2200** and/or other receives or detects input from a source external to **2240**, such as a touchpad, a touchscreen, a keyboard, a keypad, a button, a switch, a microphone or other audio input component, etc. In some embodiments, input component **2240** may include, or may be communicatively coupled to, one or more sensors, such as a motion sensor (e.g., which may be or may include a gyroscope, accelerometer, or the like), a location sensor (e.g., a Global Positioning System ("GPS")-based location sensor or some other suitable type of location sensor or location determination component), a thermometer, a barometer, and/or some other type of sensor. Output component **2250** may include a mechanism that outputs information to the operator, such as a display, a speaker, one or more light emitting diodes ("LEDs"), etc.

Communication interface **2260** may include any transceiver-like mechanism that enables device **2200** to communicate with other devices and/or systems. For example, communication interface **2260** may include an Ethernet interface, an optical interface, a coaxial interface, or the like. Communication interface **2260** may include a wireless communication device, such as an infrared ("IR") receiver, a Bluetooth® radio, or the like. The wireless communication device may be coupled to an external device, such as a remote control, a wireless keyboard, a mobile telephone, etc. In some embodiments, device **2200** may include more than one communication interface **2260**. For instance, device **2200** may include an optical interface and an Ethernet interface.

Device **2200** may perform certain operations relating to one or more processes described above. Device **2200** may perform these operations in response to processor **2220** executing software instructions stored in a computer-readable medium, such as memory **2230**. A computer-readable medium may be defined as a non-transitory memory device. A memory device may include space within a single physical memory device or spread across multiple physical

memory devices. The software instructions may be read into memory **2230** from another computer-readable medium or from another device. The software instructions stored in memory **2230** may cause processor **2220** to perform processes described herein. Alternatively, hardwired circuitry may be used in place of or in combination with software instructions to implement processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

The foregoing description of implementations provides illustration and description, but is not intended to be exhaustive or to limit the possible implementations to the precise form disclosed. Modifications and variations are possible in light of the above disclosure or may be acquired from practice of the implementations.

For example, while series of blocks and/or signals have been described above (e.g., with regard to FIGS. **1-18**), the order of the blocks and/or signals may be modified in other implementations. Further, non-dependent blocks and/or signals may be performed in parallel. Additionally, while the figures have been described in the context of particular devices performing particular acts, in practice, one or more other devices may perform some or all of these acts in lieu of, or in addition to, the above-mentioned devices.

The actual software code or specialized control hardware used to implement an embodiment is not limiting of the embodiment. Thus, the operation and behavior of the embodiment has been described without reference to the specific software code, it being understood that software and control hardware may be designed based on the description herein.

In the preceding specification, various example embodiments have been described with reference to the accompanying drawings. It will, however, be evident that various modifications and changes may be made thereto, and additional embodiments may be implemented, without departing from the broader scope of the invention as set forth in the claims that follow. The specification and drawings are accordingly to be regarded in an illustrative rather than restrictive sense.

Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of the possible implementations. In fact, many of these features may be combined in ways not specifically recited in the claims and/or disclosed in the specification. Although each dependent claim listed below may directly depend on only one other claim, the disclosure of the possible implementations includes each dependent claim in combination with every other claim in the claim set.

Further, while certain connections or devices are shown, in practice, additional, fewer, or different, connections or devices may be used. Furthermore, while various devices and networks are shown separately, in practice, the functionality of multiple devices may be performed by a single device, or the functionality of one device may be performed by multiple devices. Further, multiple ones of the illustrated networks may be included in a single network, or a particular network may include multiple networks. Further, while some devices are shown as communicating with a network, some such devices may be incorporated, in whole or in part, as a part of the network.

To the extent the aforementioned implementations collect, store, or employ personal information of individuals, groups or other entities, it should be understood that such information shall be used in accordance with all applicable laws concerning protection of personal information. Additionally,

33

the collection, storage, and use of such information can be subject to consent of the individual to such activity, for example, through well known “opt-in” or “opt-out” processes as can be appropriate for the situation and type of information. Storage and use of personal information can be in an appropriately secure manner reflective of the type of information, for example, through various access control, encryption and anonymization techniques for particularly sensitive information.

No element, act, or instruction used in the present application should be construed as critical or essential unless explicitly described as such. An instance of the use of the term “and,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Similarly, an instance of the use of the term “or,” as used herein, does not necessarily preclude the interpretation that the phrase “and/or” was intended in that instance. Also, as used herein, the article “a” is intended to include one or more items, and may be used interchangeably with the phrase “one or more.” Where only one item is intended, the terms “one,” “single,” “only,” or similar language is used. Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise.

What is claimed is:

1. A device, comprising:

one or more processors configured to:

receive an indication that a particular application, associated with the device, has been authorized by a network to use a particular set of traffic parameters;

determine, based on the authorization by the network for the particular application to use the particular set of traffic parameters, a device-level authorization for the particular application to use the particular set of traffic parameters, wherein determining the device-level authorization for the particular application to use the particular set of traffic parameters includes receiving an indication from a user of the device that the particular application is authorized to use the particular set of traffic parameters;

selectively apply the set of traffic parameters to traffic associated with the particular application, wherein selectively applying the set of traffic parameters includes:

applying the particular set of traffic parameters to traffic associated with the particular application when it is determined that the particular application has device-level authorization to use the particular set of traffic parameters, and

refraining from applying the particular set of traffic parameters to the traffic associated with the particular application when it is determined that the particular application does not have device-level authorization to use the particular set of traffic parameters; and

output the traffic via the network after applying the particular set of traffic parameters or refraining from applying the particular set of traffic parameters.

2. The device of claim 1,

wherein the device presents a user interface that includes:

the first option to authorize the particular application to use the particular set of traffic parameters, and

a second option to forgo authorizing the particular application to use the particular set of traffic parameters, and

wherein the indication, that the particular application is authorized to use the particular set of traffic parameters,

34

is received via a user selection of the first option in the user interface presented by the device.

3. The device of claim 1, wherein the device presents a user interface based on receiving the indication that the particular application has been authorized by the network to use the particular set of traffic parameters, wherein the indication, that the particular application is authorized to use the particular set of traffic parameters, is received via the user interface.

4. The device of claim 1, wherein the particular application is a first application, wherein the one or more processors are further configured to:

determine that a second application, associated with the device, is not authorized by the network to use the particular set of traffic parameters; and

output traffic, associated with the second application, to the network without applying the particular set of traffic parameters, based on the determination that the second application is not authorized by the network to use the particular set of traffic parameters.

5. A device, comprising:

one or more processors configured to:

receive an indication that a particular application, associated with the device, has been authorized by a network to use a particular set of traffic parameters;

determine, based on the authorization by the network for the particular application to use the particular set of traffic parameters, a device-level authorization for the particular application to use the particular set of traffic parameters;

selectively apply the set of traffic parameters to traffic associated with the particular application, wherein selectively applying the set of traffic parameters includes:

applying the particular set of traffic parameters to traffic associated with the particular application when it is determined that the particular application has device-level authorization to use the particular set of traffic parameters, and

refraining from applying the particular set of traffic parameters to the traffic associated with the particular application when it is determined that the particular application does not have device-level authorization to use the particular set of traffic parameters;

identify a plurality of applications, associated with the device, that are authorized by the network to use the particular set of traffic parameters;

receive an indication of which applications, of the plurality of applications, are authorized by a user of the device to use the particular set of traffic parameters, wherein the indication specifies a first set of the plurality of applications and omits a second set of the plurality of applications; and

output the traffic, associated with the particular application, via the network after applying the particular set of traffic parameters or refraining from applying the particular set of traffic parameters, wherein outputting the traffic includes applying the particular set of traffic parameters to traffic associated with the first set of applications without applying the particular set of traffic parameters to traffic associated with the second set of applications.

6. The device of claim 5, wherein applying the particular set of traffic parameters comprises including an indicator,

35

associated with the particular set of traffic parameters, in header information of the traffic associated with the particular application.

7. The device of claim 5, wherein the one or more processors are further configured to:

determine that a second application, associated with the device, is not authorized by the network to use the particular set of traffic parameters; and

output traffic, associated with the second application, to the network without applying the particular set of traffic parameters, based on the determination that the second application is not authorized by the network to use the particular set of traffic parameters.

8. A non-transitory computer-readable medium, storing a plurality of processor-executable instructions to:

receive an indication that a particular application, associated with a device, has been authorized by a network to use a particular set of traffic parameters;

determine, based on the authorization by the network for the particular application to use the particular set of traffic parameters, a device-level authorization for the particular application to use the particular set of traffic parameters, wherein determining the device-level authorization for the particular application to use the particular set of traffic parameters includes receiving an indication from a user of the device that the particular application is authorized to use the particular set of traffic parameters;

selectively apply the set of traffic parameters to traffic associated with the particular application, wherein selectively applying the set of traffic parameters includes:

applying the particular set of traffic parameters to traffic associated with the particular application when it is determined that the particular application has device-level authorization to use the particular set of traffic parameters, and

refraining from applying the particular set of traffic parameters to the traffic associated with the particular application when it is determined that the particular application does not have device-level authorization to use the particular set of traffic parameters; and

output the traffic via the network after applying the particular set of traffic parameters or refraining from applying the particular set of traffic parameters.

9. The non-transitory computer-readable medium of claim 8,

wherein the device presents a user interface that includes:

the first option to authorize the particular application to use the particular set of traffic parameters, and

a second option to forgo authorizing the particular application to use the particular set of traffic parameters, and

wherein the indication, that the particular application is authorized to use the particular set of traffic parameters, is received via a user selection of the first option in the user interface presented by the device.

10. The non-transitory computer-readable medium of claim 8, wherein the device presents a user interface based on receiving the indication that the particular application has been authorized by the network to use the particular set of traffic parameters, wherein the indication, that the particular application is authorized to use the particular set of traffic parameters, is received via the user interface.

11. The non-transitory computer-readable medium of claim 8, wherein the particular application is a first appli-

36

cation, wherein the plurality of processor-executable instructions further include processor-executable instructions to:

determine that a second application, associated with the device, is not authorized by the network to use the particular set of traffic parameters; and

output traffic, associated with the second application, to the network without applying the particular set of traffic parameters, based on the determination that the second application is not authorized by the network to use the particular set of traffic parameters.

12. A non-transitory computer-readable medium, storing a plurality of processor-executable instructions to:

receive an indication that a particular application, associated with the device, has been authorized by a network to use a particular set of traffic parameters;

determine, based on the authorization by the network for the particular application to use the particular set of traffic parameters, a device-level authorization for the particular application to use the particular set of traffic parameters;

selectively apply the set of traffic parameters to traffic associated with the particular application, wherein selectively applying the set of traffic parameters includes:

applying the particular set of traffic parameters to traffic associated with the particular application when it is determined that the particular application has device-level authorization to use the particular set of traffic parameters, and

refraining from applying the particular set of traffic parameters to the traffic associated with the particular application when it is determined that the particular application does not have device-level authorization to use the particular set of traffic parameters;

identify a plurality of applications, associated with the device, that are authorized by the network to use the particular set of traffic parameters;

receive an indication of which applications, of the plurality of applications, are authorized by a user of the device to use the particular set of traffic parameters, wherein the indication specifies a first set of the plurality of applications and omits a second set of the plurality of applications; and

output the traffic, associated with the particular application, via the network after applying the particular set of traffic parameters or refraining from applying the particular set of traffic parameters, wherein outputting the traffic includes applying the particular set of traffic parameters to traffic associated with the first set of applications without applying the particular set of traffic parameters to traffic associated with the first set of applications.

13. The non-transitory computer-readable medium of claim 12, wherein applying the particular set of traffic parameters comprises including an indicator, associated with the particular set of traffic parameters, in header information of the traffic associated with the particular application.

14. The non-transitory computer-readable medium of claim 12, wherein the plurality of processor-executable instructions further include processor-executable instructions to:

determine that a second application, associated with the device, is not authorized by the network to use the particular set of traffic parameters; and

37

output traffic, associated with the second application, to the network without applying the particular set of traffic parameters, based on the determination that the second application is not authorized by the network to use the particular set of traffic parameters.

15. A method, comprising:

receiving an indication that a particular application, associated with a device, has been authorized by a network to use a particular set of traffic parameters;

determining, based on the authorization by the network for the particular application to use the particular set of traffic parameters, a device-level authorization for the particular application to use the particular set of traffic parameters, wherein determining the device-level authorization for the particular application to use the particular set of traffic parameters includes receiving an indication from a user of the device that the particular application is authorized to use the particular set of traffic parameters;

selectively applying the set of traffic parameters to traffic associated with the particular application, wherein selectively applying the set of traffic parameters includes:

applying the particular set of traffic parameters to traffic associated with the particular application when it is determined that the particular application has device-level authorization to use the particular set of traffic parameters, and

refraining from applying the particular set of traffic parameters to the traffic associated with the particular application when it is determined that the particular application does not have device-level authorization to use the particular set of traffic parameters; and

outputting the traffic via the network after applying the particular set of traffic parameters or refraining from applying the particular set of traffic parameters.

16. The method of claim 15,

wherein the device presents a user interface that includes: the first option to authorize the particular application to use the particular set of traffic parameters, and a second option to forgo authorizing the particular application to use the particular set of traffic parameters, and

wherein the indication, that the particular application is authorized to use the particular set of traffic parameters, is received via a user selection of the first option in the user interface presented by the device.

17. The method of claim 16, wherein the particular application is a first application, the method further comprising:

determining that a second application, associated with the device, is not authorized by the network to use the particular set of traffic parameters; and

outputting traffic, associated with the second application, to the network without applying the particular set of traffic parameters, based on the determination that the second application is not authorized by the network to use the particular set of traffic parameters.

18. A method, comprising:

38

receiving an indication that a particular application, associated with the device, has been authorized by a network to use a particular set of traffic parameters;

determining, based on the authorization by the network for the particular application to use the particular set of traffic parameters, a device-level authorization for the particular application to use the particular set of traffic parameters;

selectively applying the set of traffic parameters to traffic associated with the particular application, wherein selectively applying the set of traffic parameters includes:

applying the particular set of traffic parameters to traffic associated with the particular application when it is determined that the particular application has device-level authorization to use the particular set of traffic parameters, and

refraining from applying the particular set of traffic parameters to the traffic associated with the particular application when it is determined that the particular application does not have device-level authorization to use the particular set of traffic parameters;

identifying a plurality of applications, associated with the device, that are authorized by the network to use the particular set of traffic parameters;

receiving an indication of which applications, of the plurality of applications, are authorized by a user of the device to use the particular set of traffic parameters, wherein the indication specifies a first set of the plurality of applications and omits a second set of the plurality of applications; and

outputting the traffic, associated with the particular application, via the network after applying the particular set of traffic parameters or refraining from applying the particular set of traffic parameters, wherein outputting the traffic includes applying the particular set of traffic parameters to traffic associated with the first set of applications without applying the particular set of traffic parameters to traffic associated with the first set of applications.

19. The method of claim 18, wherein applying the particular set of traffic parameters comprises including an indicator, associated with the particular set of traffic parameters, in header information of the traffic associated with the particular application.

20. The method of claim 18, wherein the particular application is a first application, the method further comprising:

determining that a second application, associated with the device, is not authorized by the network to use the particular set of traffic parameters; and

outputting traffic, associated with the second application, to the network without applying the particular set of traffic parameters, based on the determination that the second application is not authorized by the network to use the particular set of traffic parameters.

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